Lab 8: Real-Time Data Transformation with Spark Streaming on AWS Glue

Lab summary

In this lab, you will build a **real-time analytics pipeline** for telecom network metrics.

Streaming data will be ingested through **Amazon Kinesis**, processed using **AWS Glue Spark Streaming**, stored in **Amazon S3**, queried using **Amazon Athena**, and visualized via a **Streamlit dashboard deployed on Amazon ECS**.

The goal is to compute near real-time metrics like:

- Average signal strength by operator
- Average GPS precision by provider
- Status counts by postal code

Prerequisites to complete lab

- AWS account with permissions for Glue, Kinesis, S3, Athena, ECS, IAM
- IAM role for Glue with required permissions
- Local machine with:
  - Python 3 installed
  - Docker installed and running
  - AWS CLI configured
- Source code files:
  - Spark streaming script
  - Kinesis producer script
  - Mobile logs CSV dataset

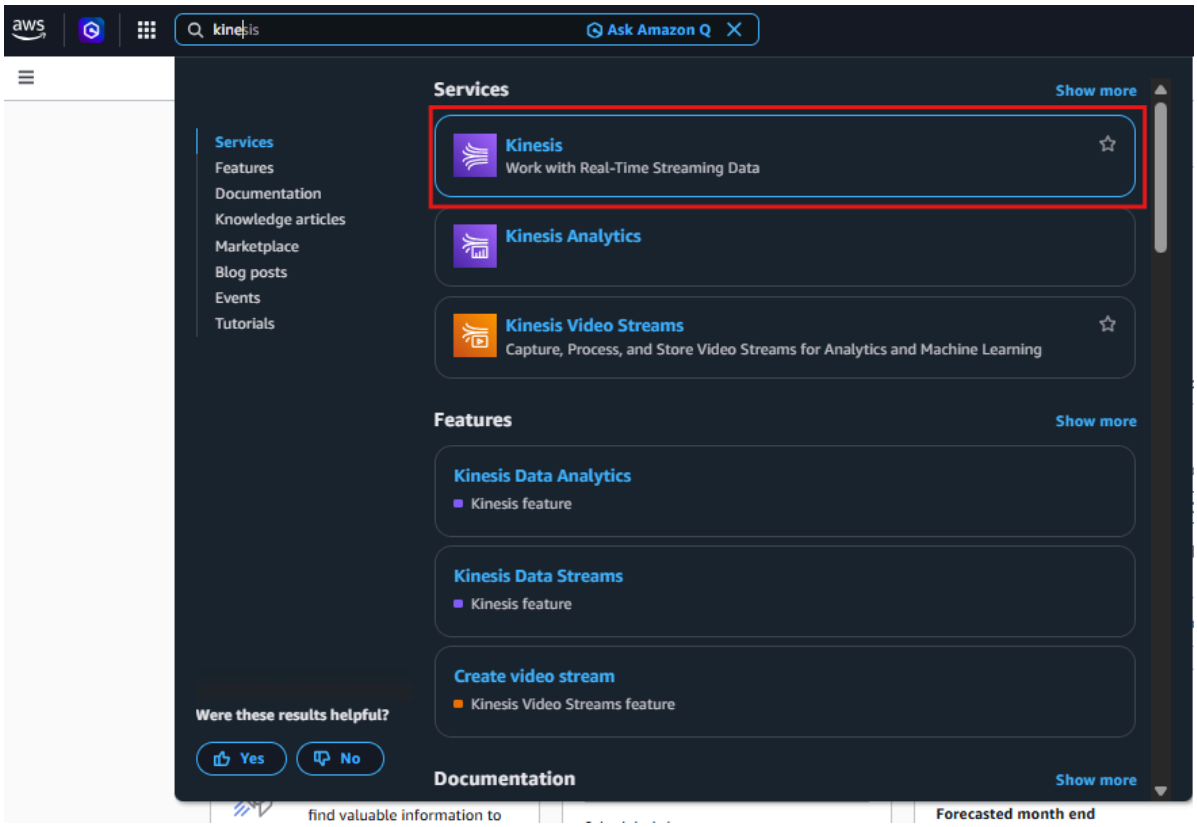
- Streamlit Docker project

Steps to complete lab

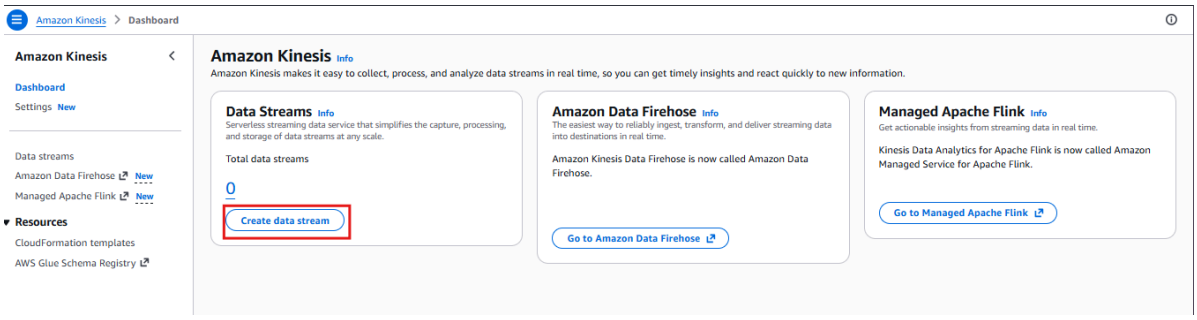
Task 1 — Create Kinesis Data Stream

Step — Open Kinesis

1. Go to AWS Console → Search **Kinesis**

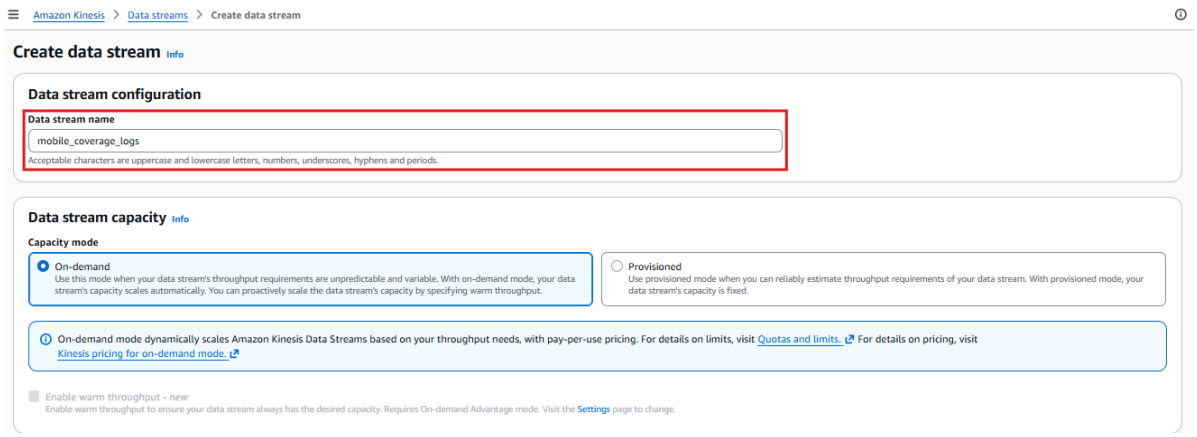


2. Click **Create Data Stream**

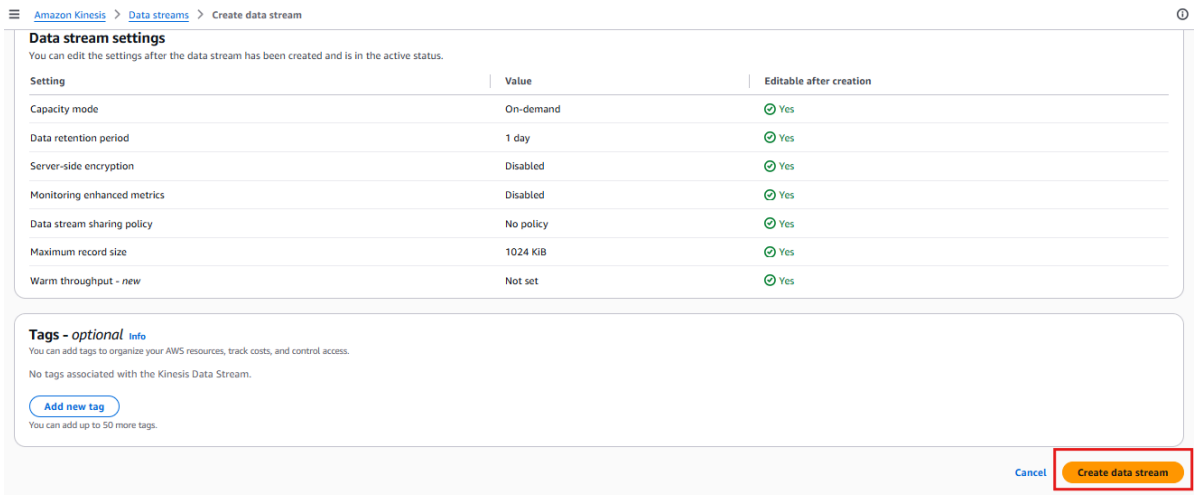


Step — Configure Stream

1. Name → `MobileCoverageLogs`

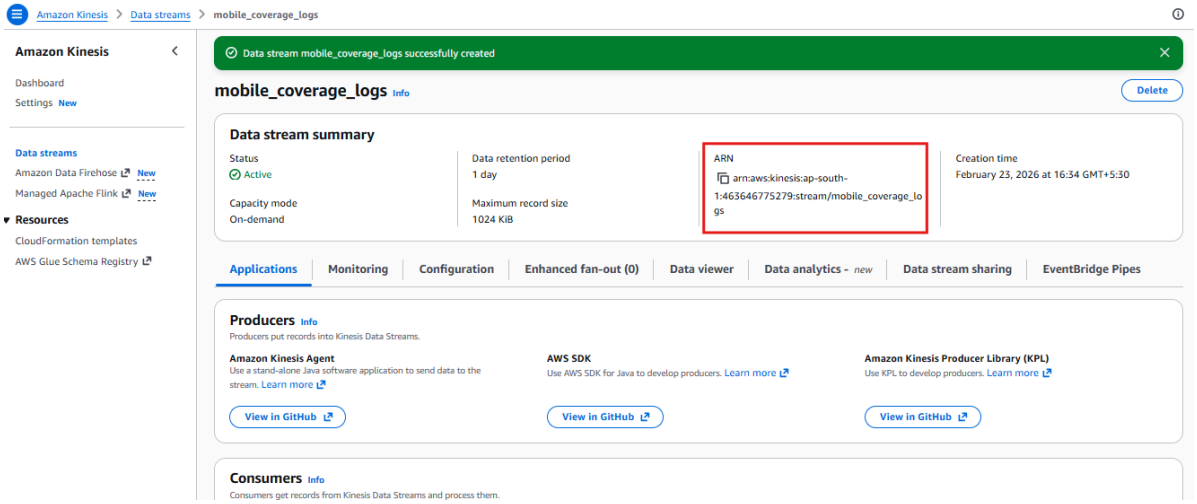


2. Leave everything else default
3. Click **Create**



Step — Copy Stream ARN

1. Open stream details
2. Copy ARN

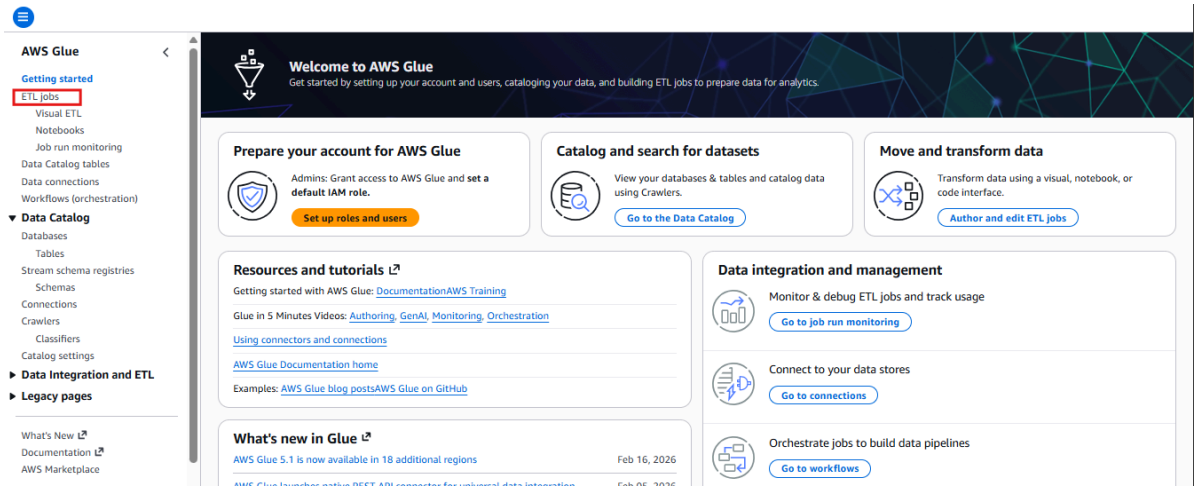


3. Paste into Spark script

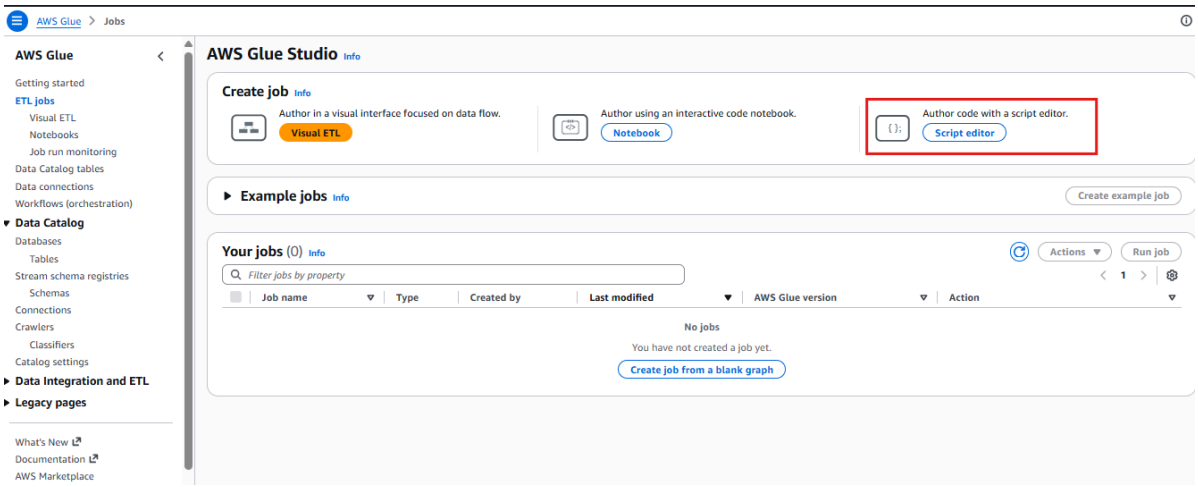
Task 2 — Configure Spark Streaming Job in Glue

Step — Open Glue

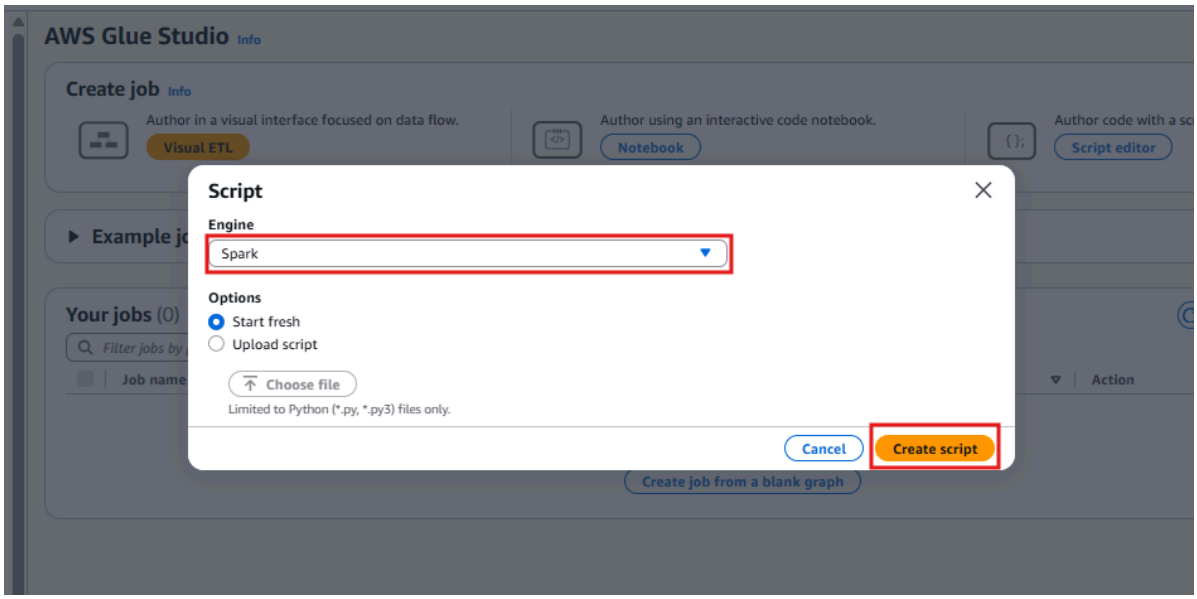
1. Go to AWS Glue Console



2. Click **Script Editor**

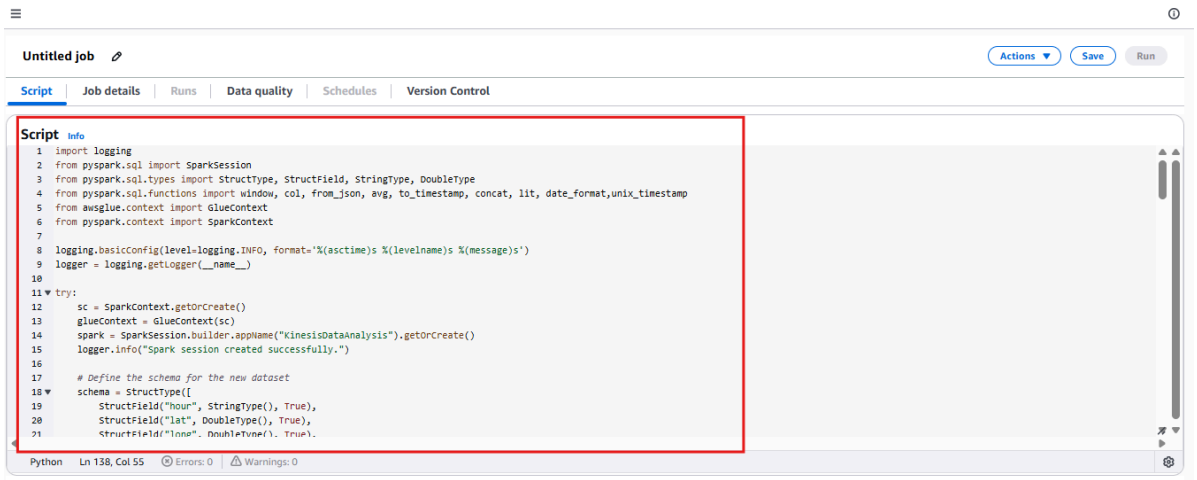


3. Select **Spark** —→ Click **create script**



Step — Paste Script

1. Copy *mobile_coverage_streaming.py* script
2. Paste into pySpark editor

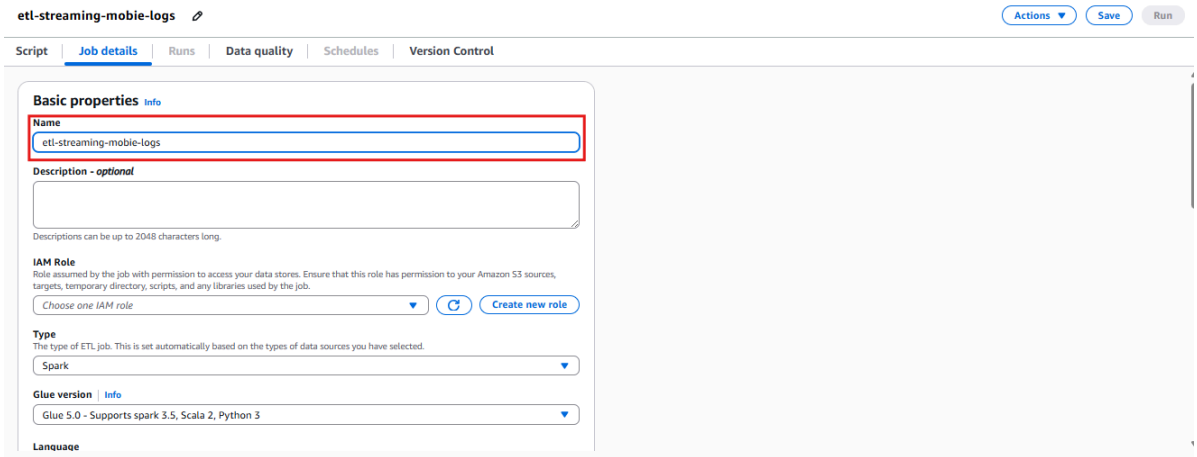


Step — Update Config

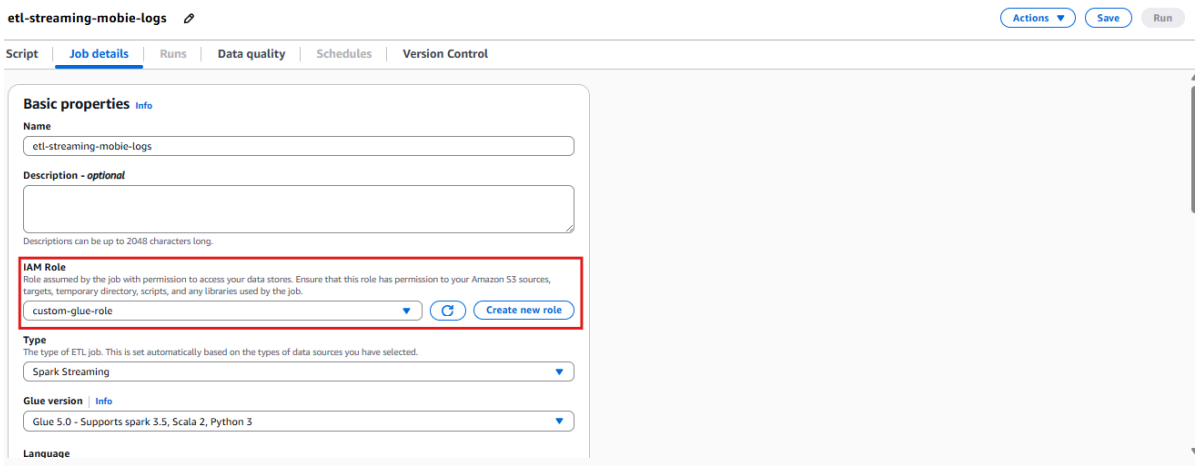
1. Replace Kinesis ARN
2. Update S3 bucket name

Step — Configure Job

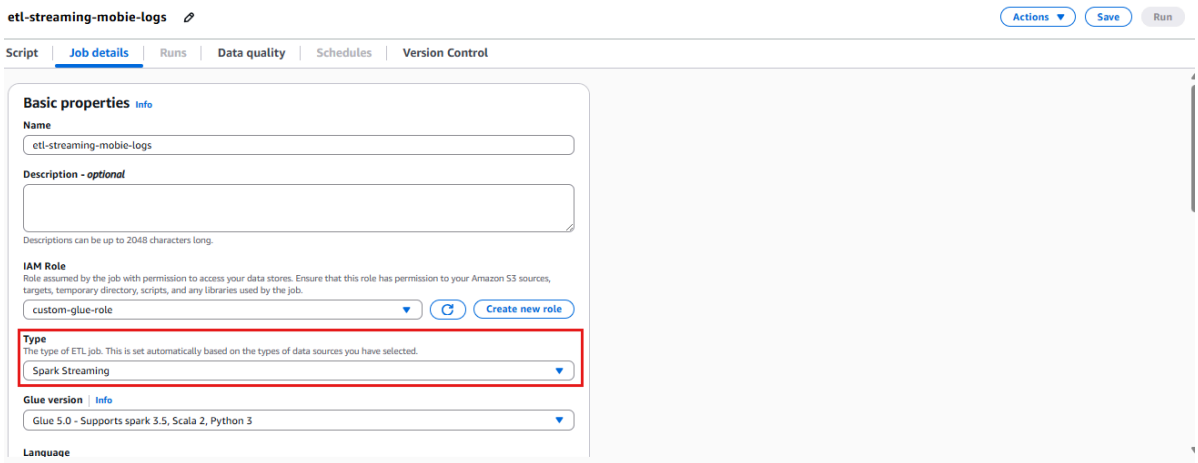
- Name the job



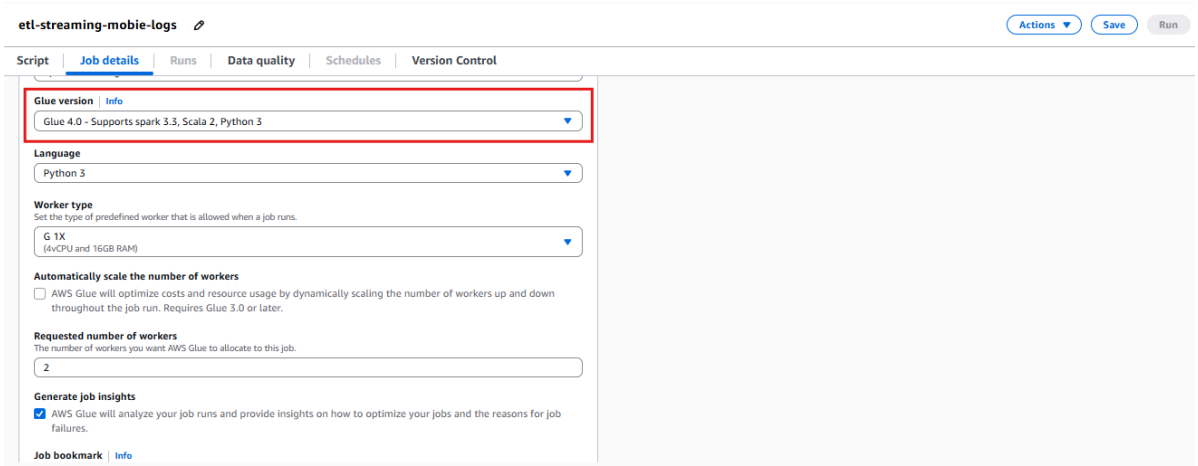
- IAM role → Custom Glue Role



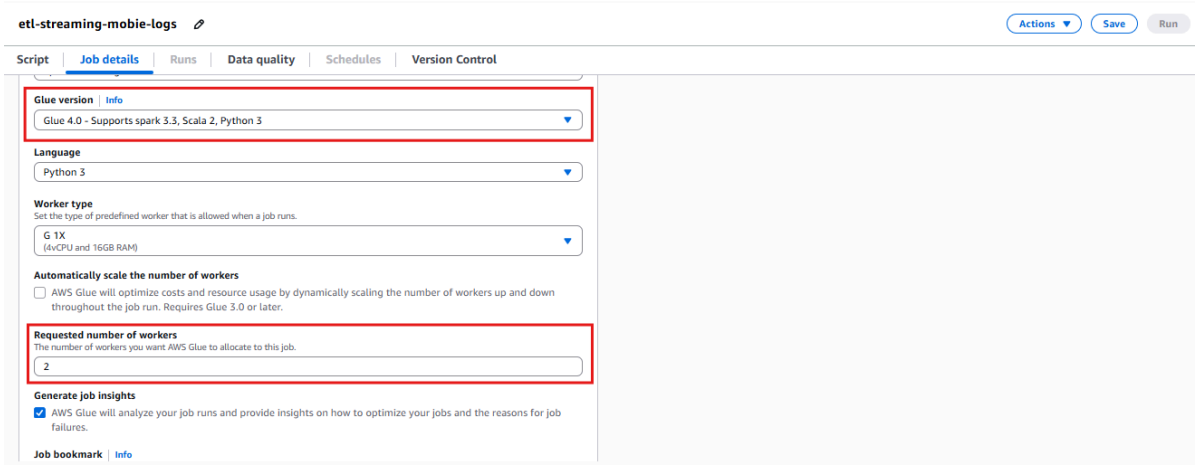
- Job type → Spark Streaming



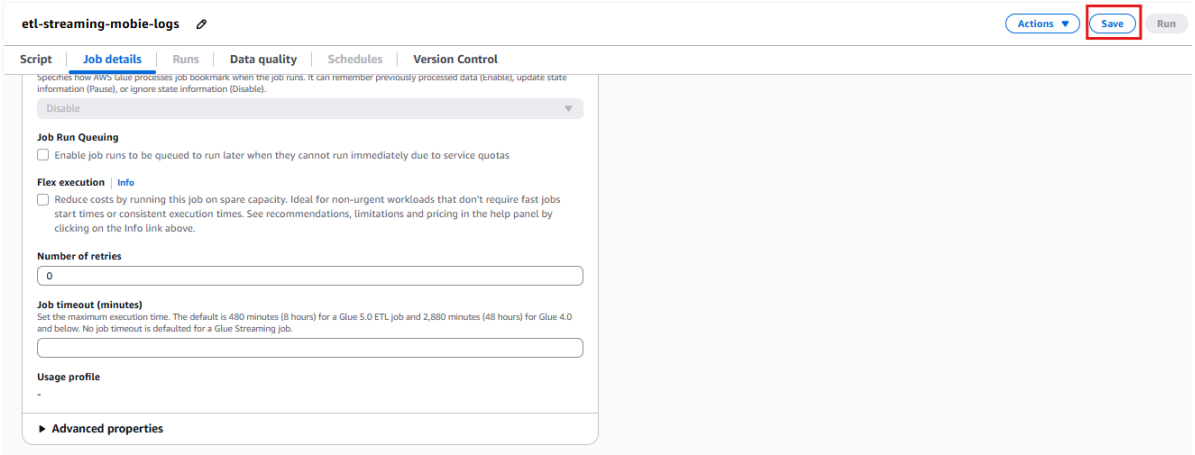
- Glue version → 4.0



- Number of Workers → 2

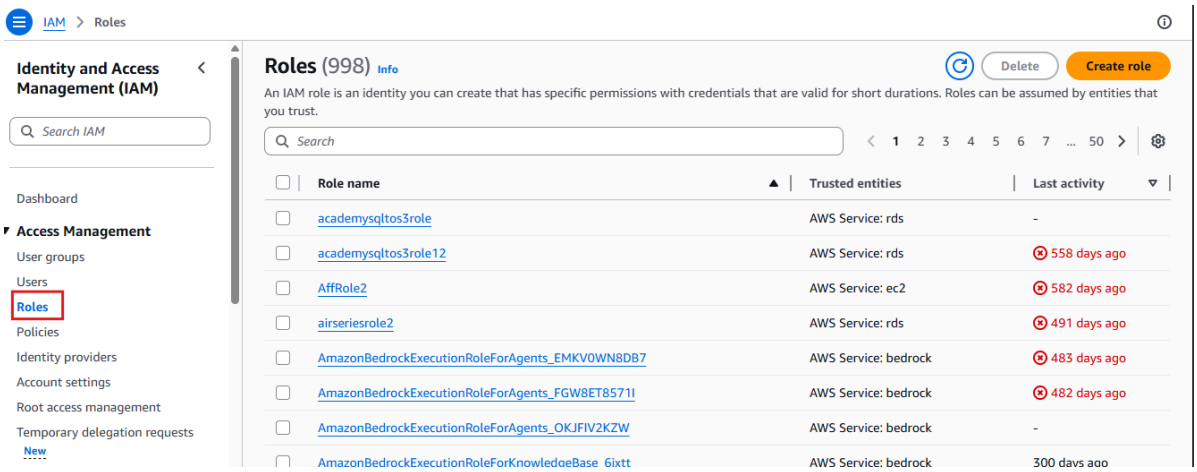


- Click on save

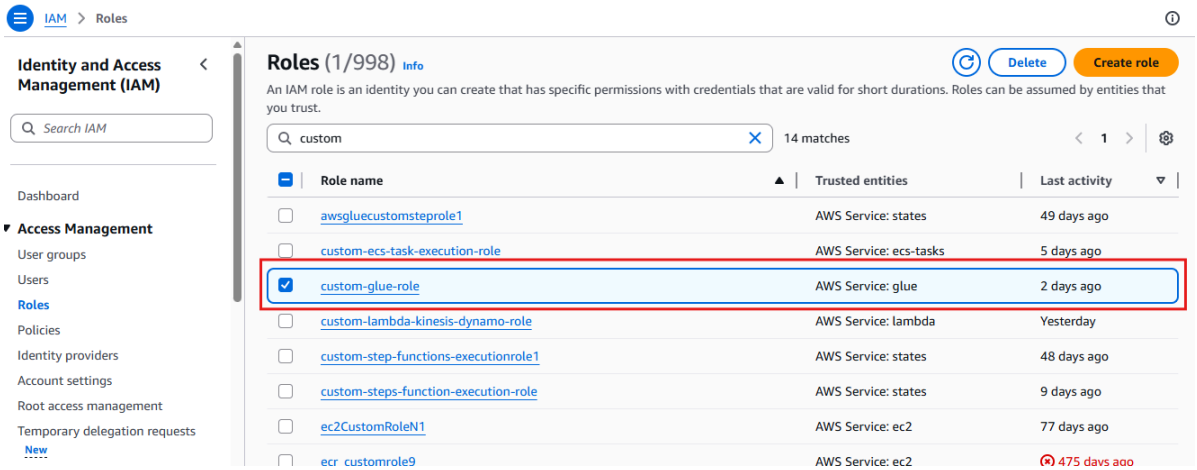


Step — Add Permission

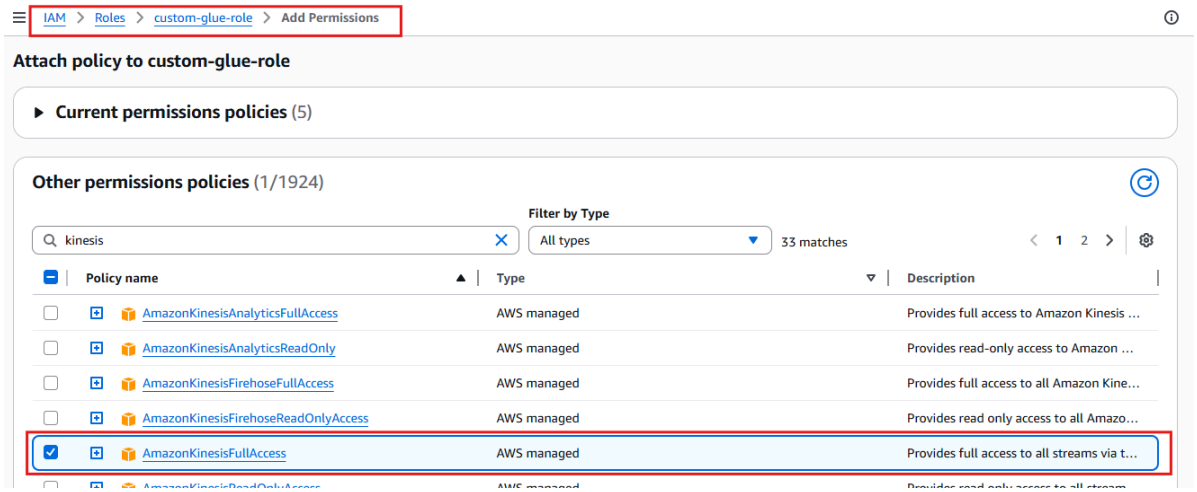
1. Go to IAM → Roles



2. Select Glue role



3. Attach policy **Kinesis Full Access**



4. Click —→ Add permission

Task 3 — Run Kinesis Producer

Step — Open terminal

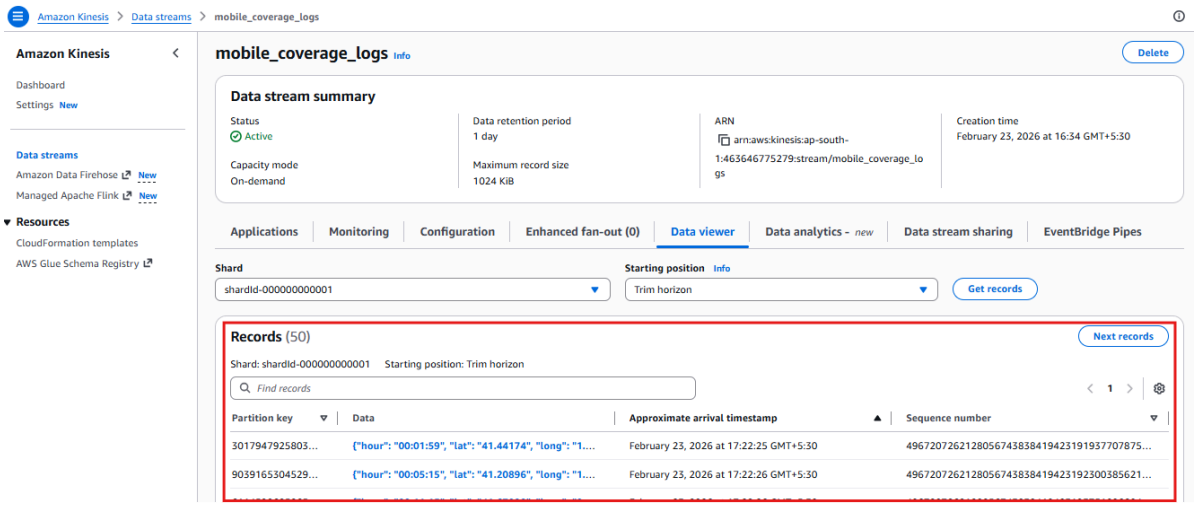
Navigate to project directory

Step — Run script

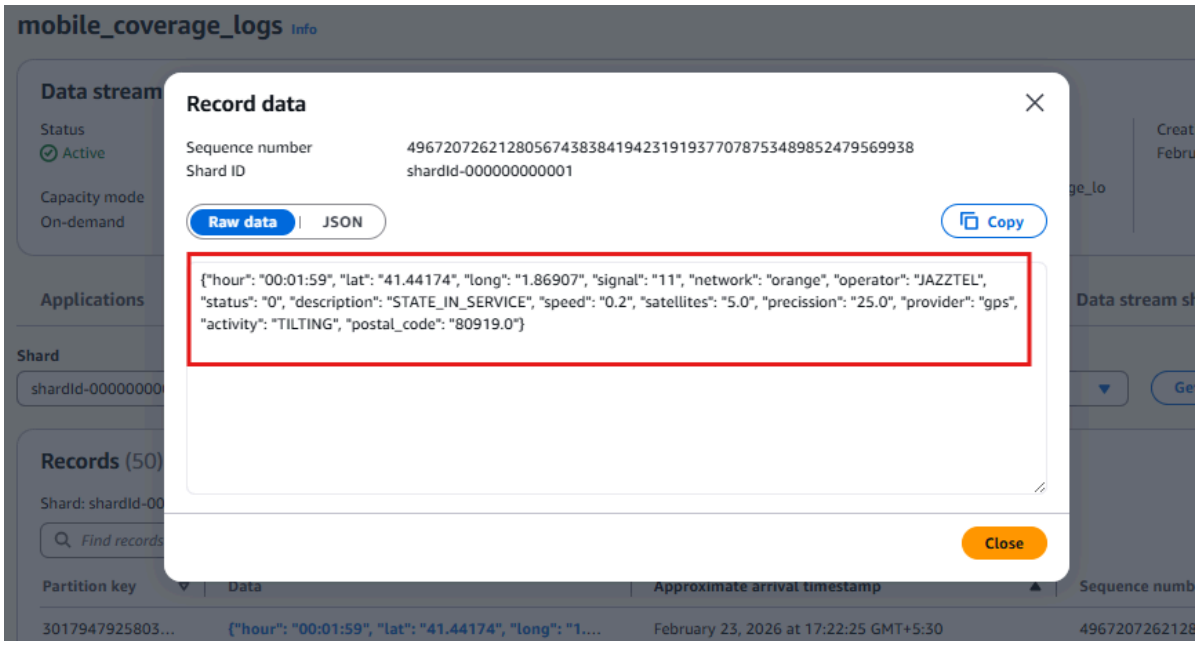
```
python3 kinesis_producer.py
```

Step — Verify ingestion

1. Open Kinesis → Data Viewer

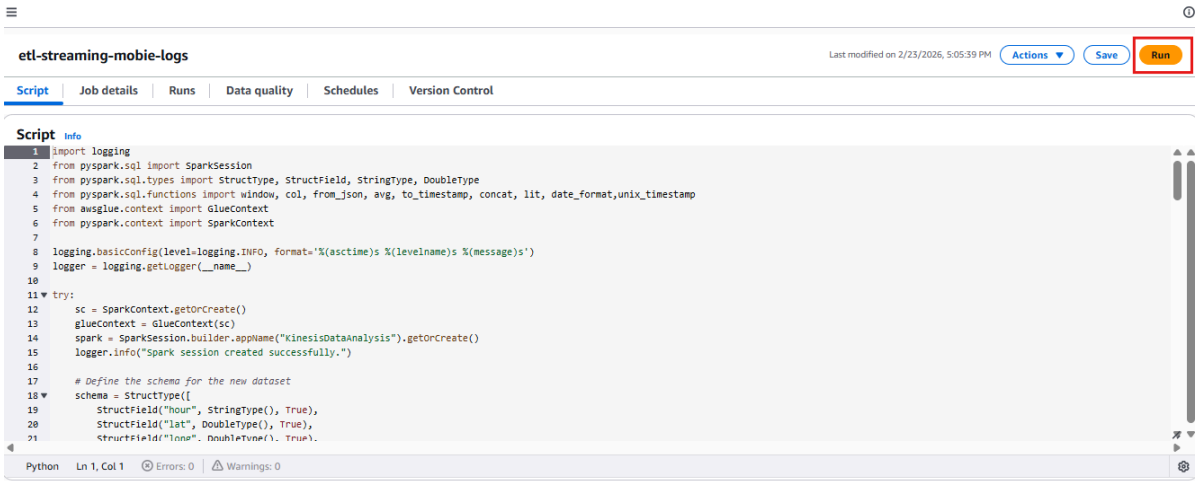


2. Check records

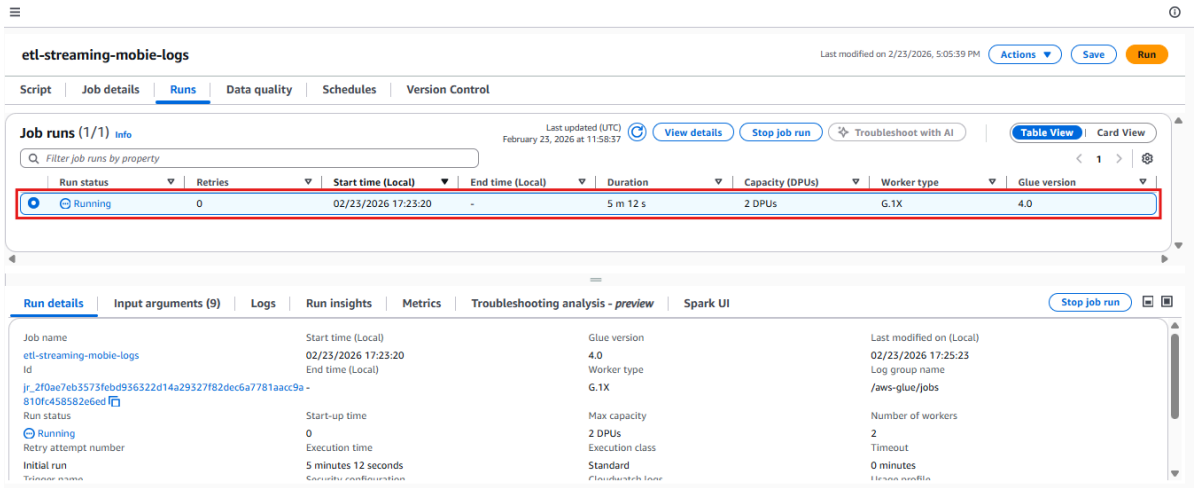


Task 4 — Run Glue Streaming Job

1. Go to Glue job —→ Click **Run**

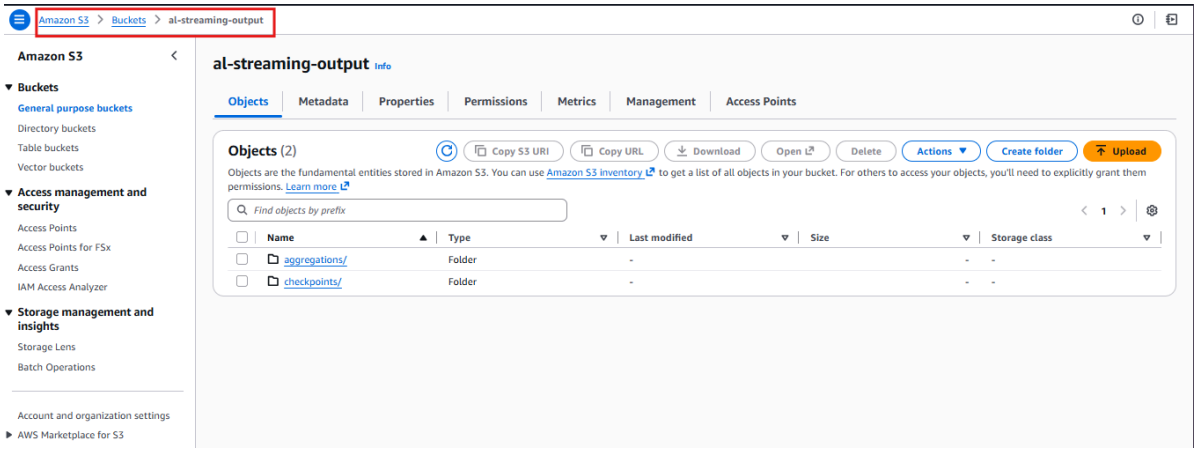


2. Monitor status

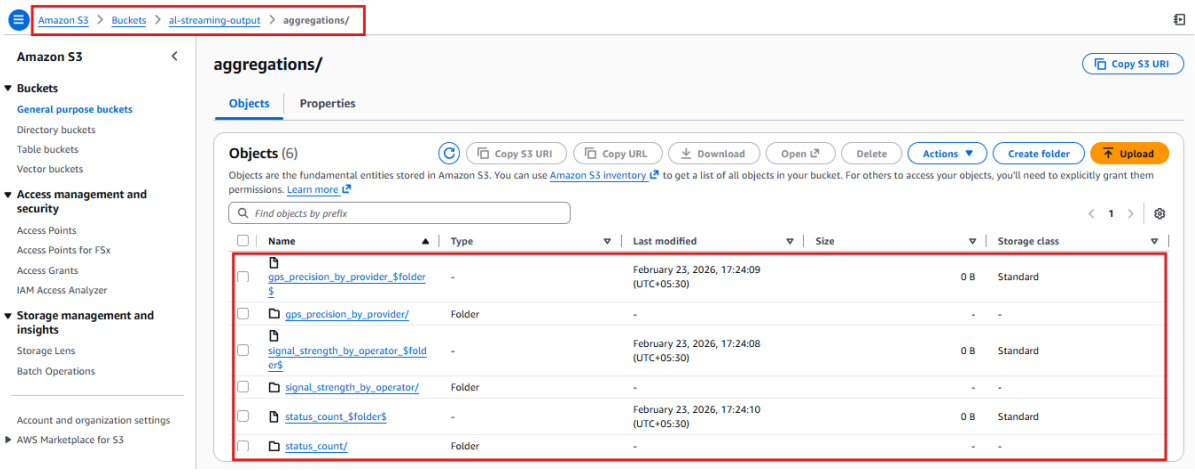


Task 5 — Verify Output in S3

1. Open S3 bucket



2. Check folder `aggregations`



You should see:

- GPS precision by provider
- Signal strength by operator
- Status count

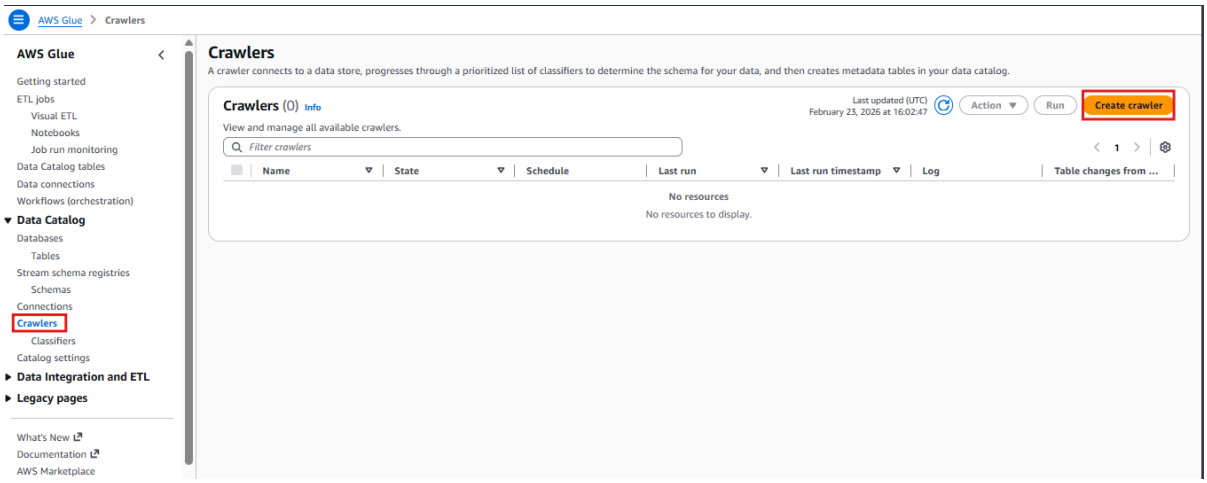
Files will be:

- Partitioned
- Snappy compressed parquet

Task 6 — Create Glue Crawlers

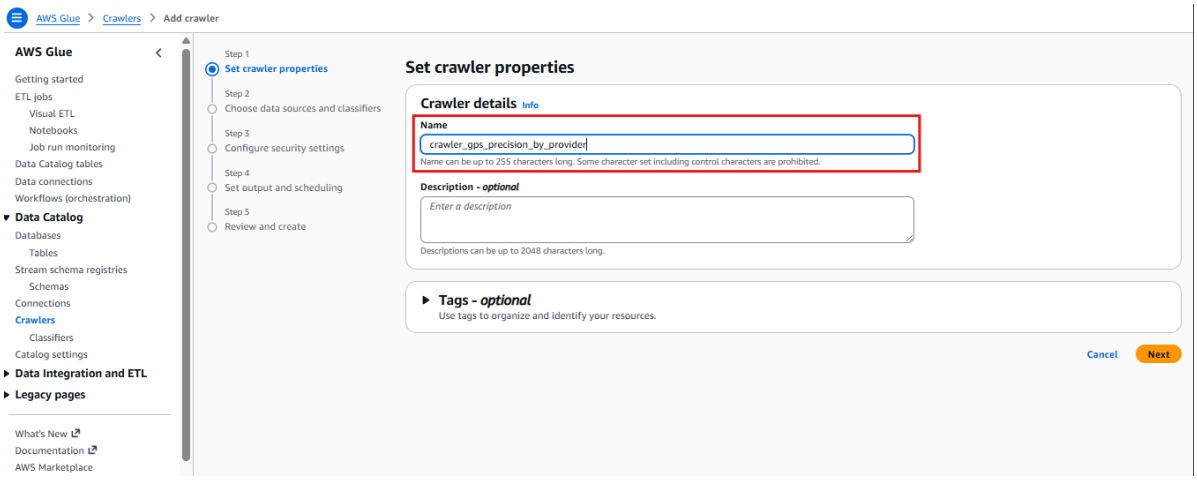
Step — Open Crawlers

Glue → Crawlers → Create

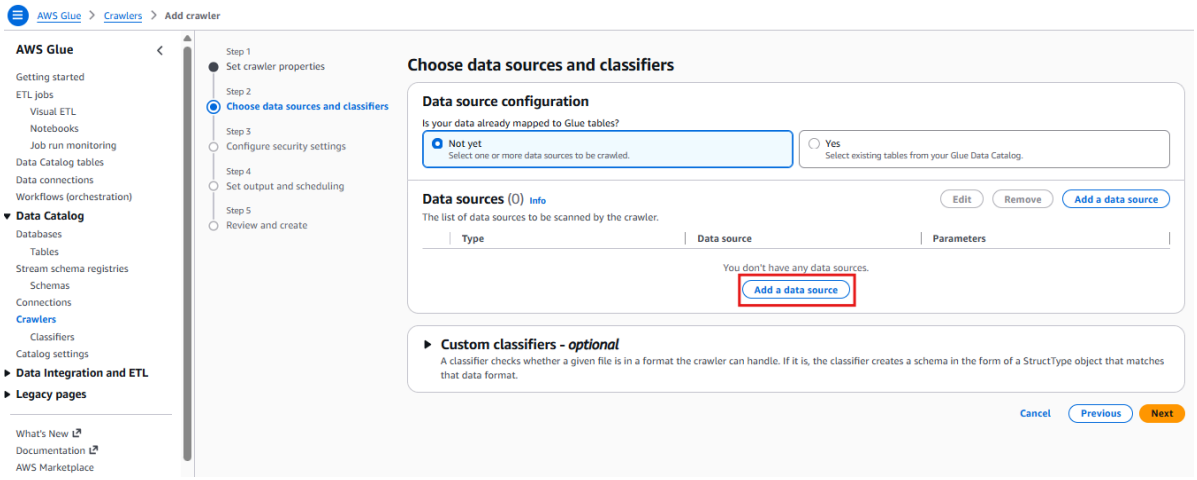


Config

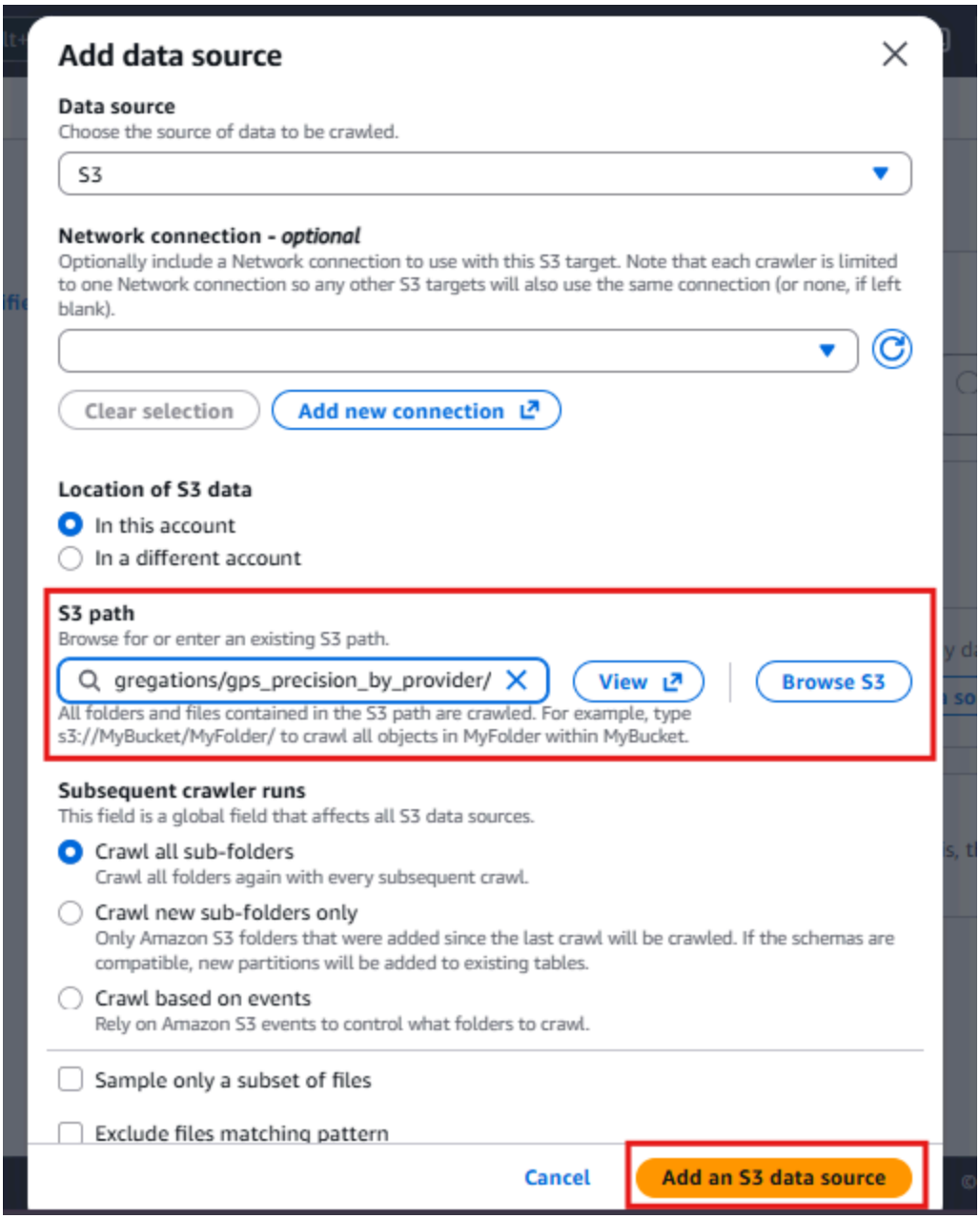
- Give a name



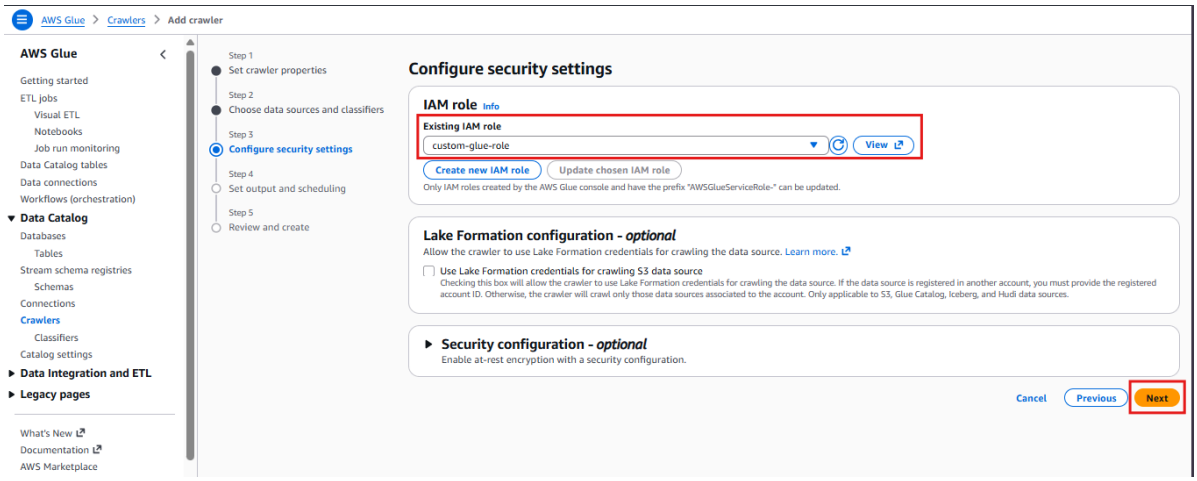
- Click Next —→ Add data source



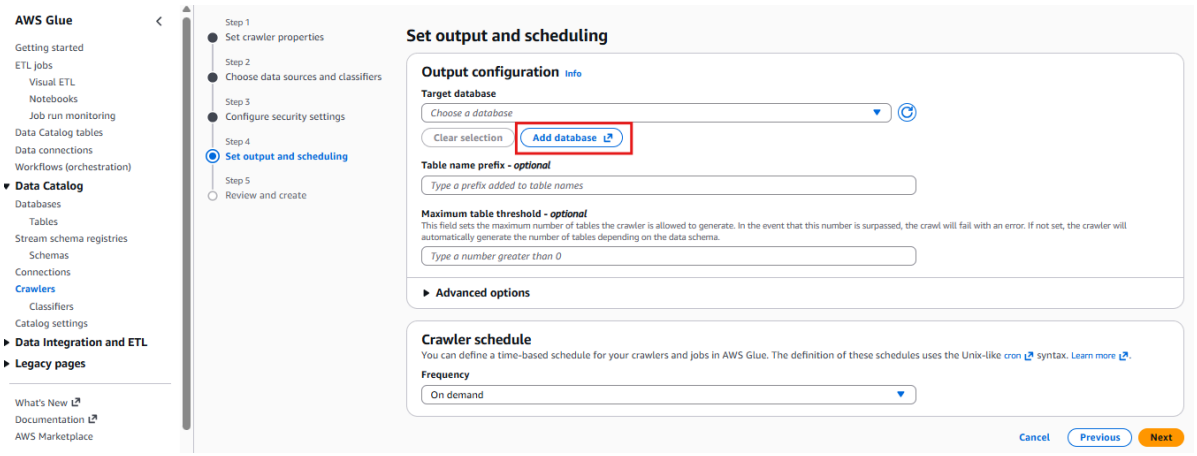
- Data source → S3 path



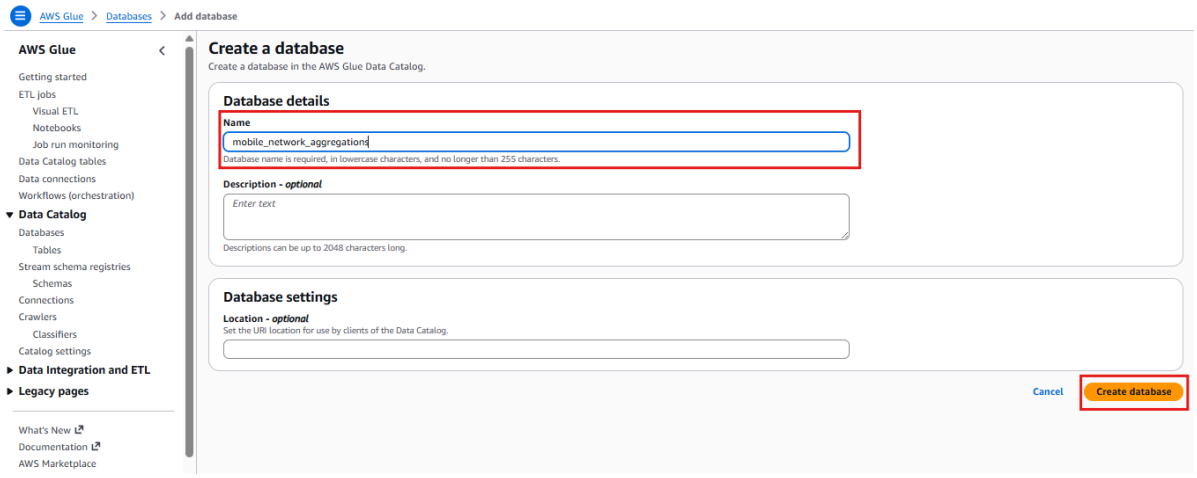
- IAM role → Glue role



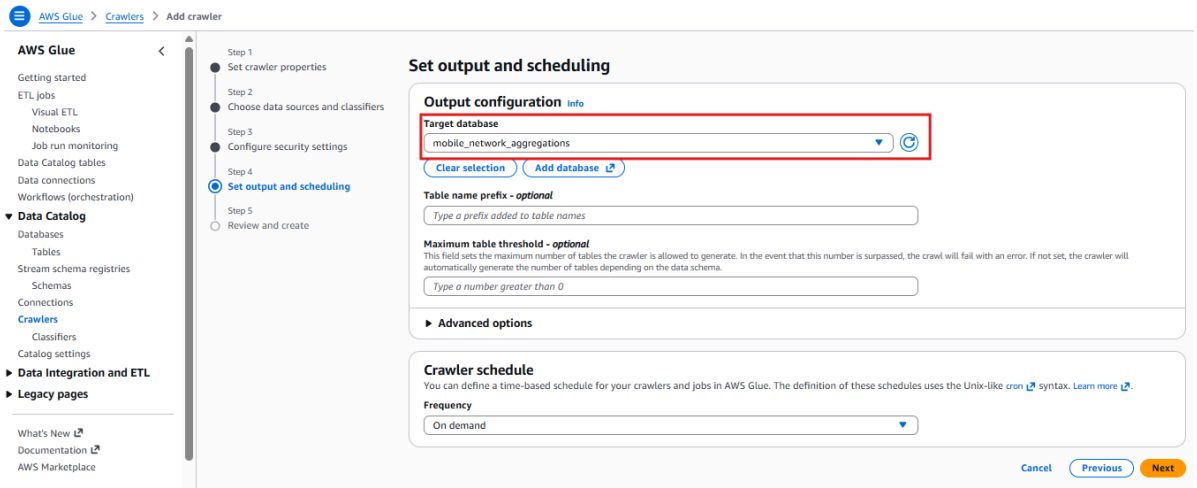
- Click Add database



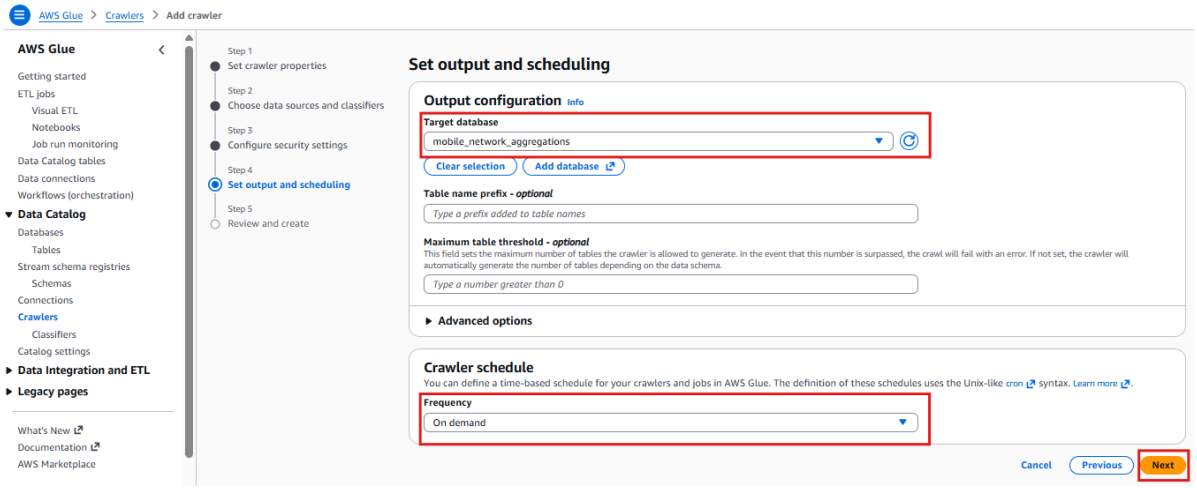
- Database name → `mobile_network_aggregations` —→ click create database



- Navigate back to crawler page —→ Select database just created



- Schedule → On demand —→ Click next



- Click Create crawler

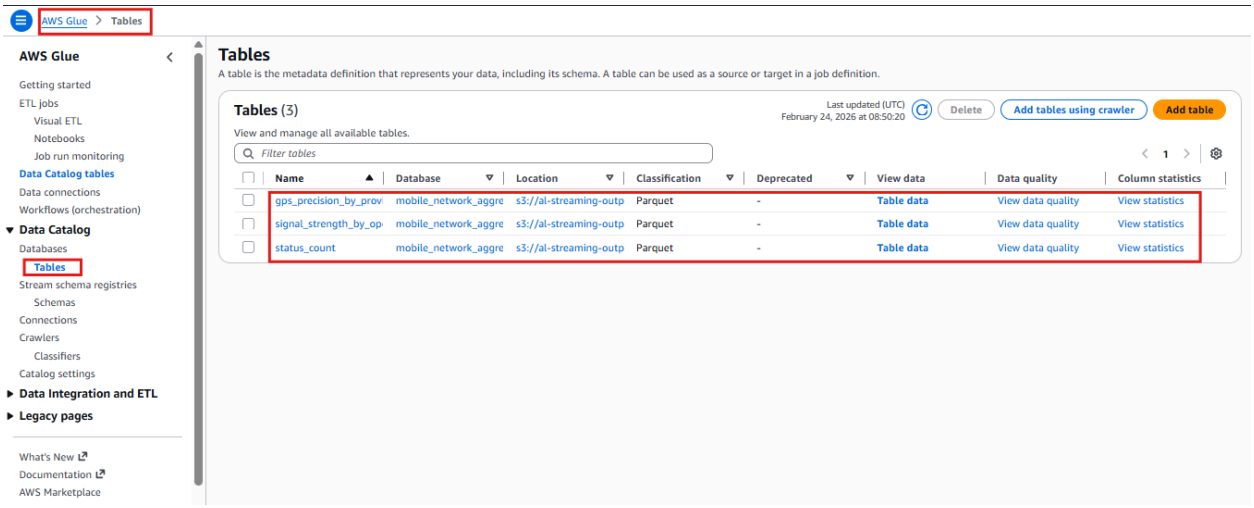
Step — Create crawler for each dataset

Repeat for all 3 datasets (follow same steps)

Step — Run crawlers

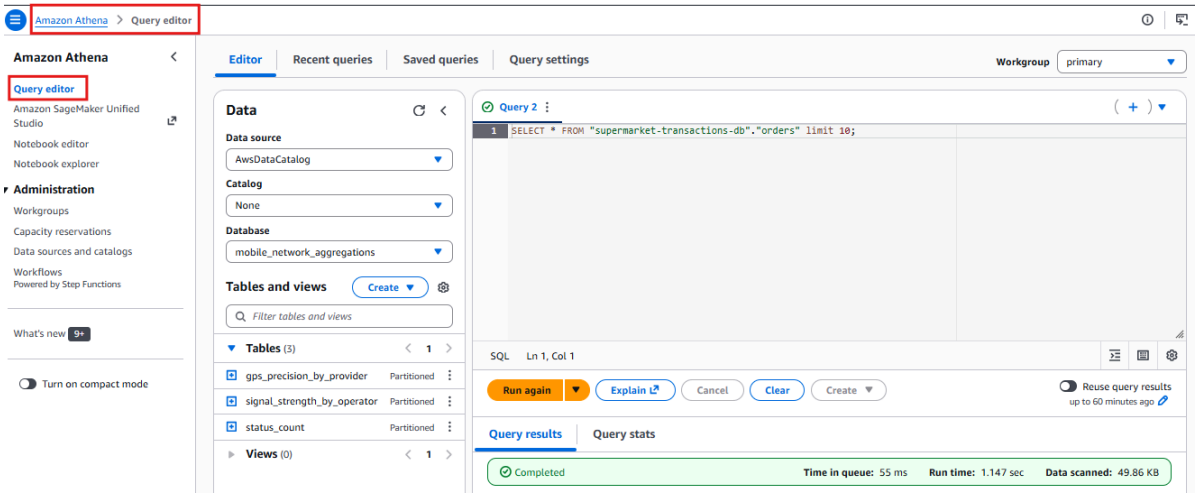
Step — Verify tables

Glue → Tables

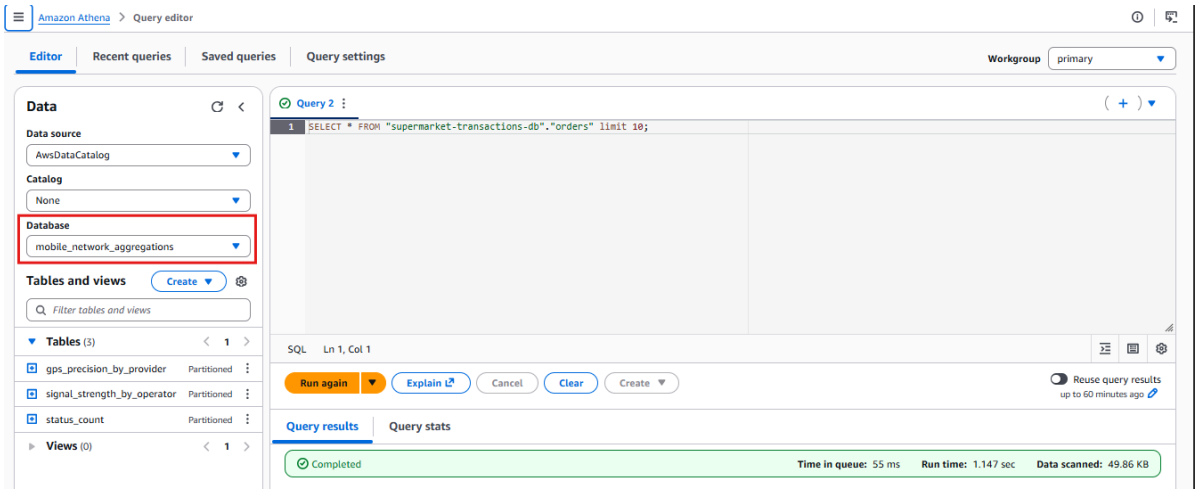


Task 7 — Query Data in Athena

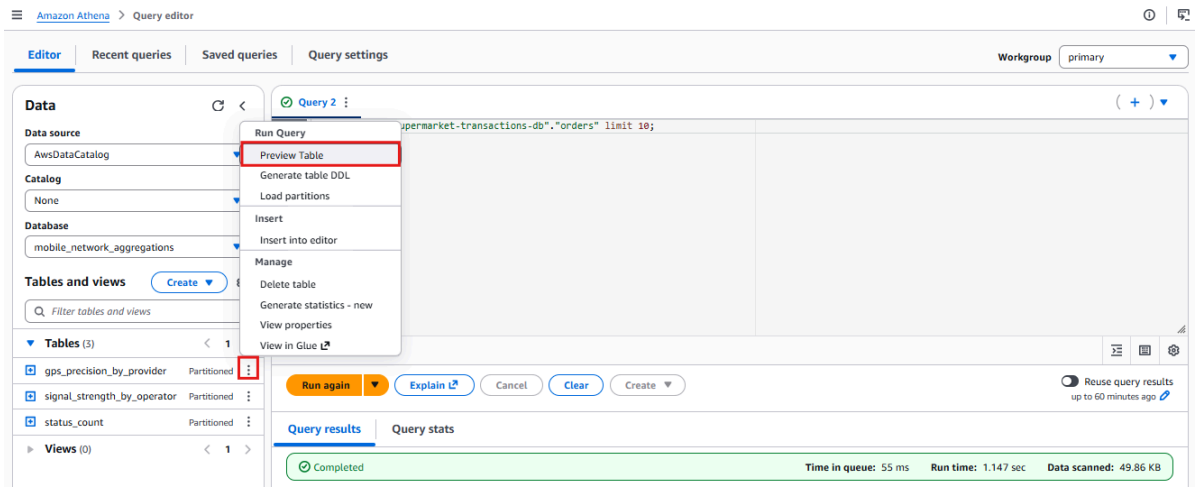
1. Open Athena —→ Query editor



2. Select database

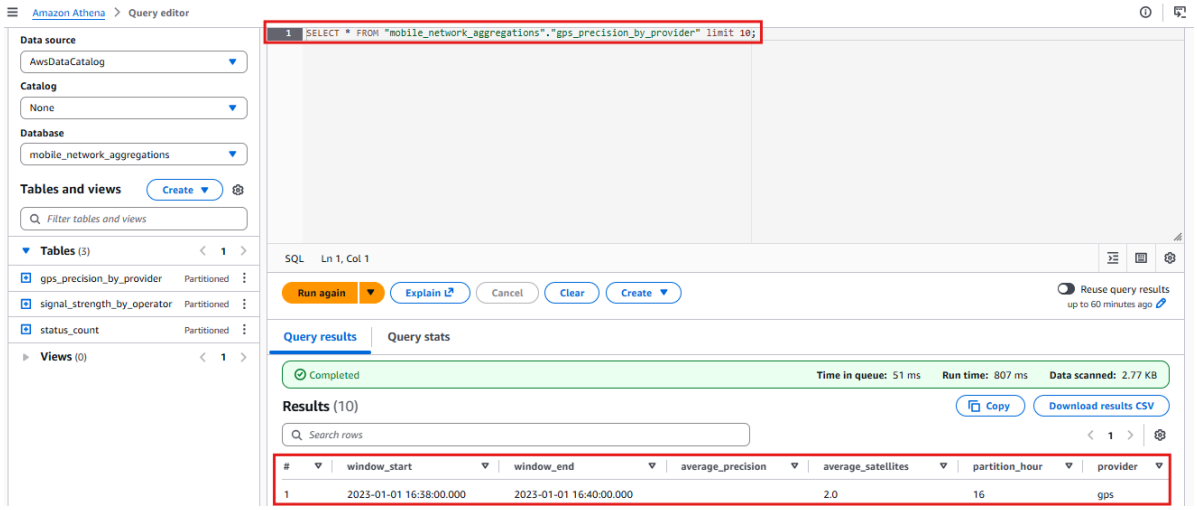


3. Run preview queries



Example:

```
SELECT * FROM gps_precision_by_provider LIMIT 10;
```



Task 8 — Run Streamlit App Locally

Step — Build Docker image

```
docker build -t mobile-signal-app .
```

Step — Run container

```
docker run -p 8501:8501 mobile-signal-app
```

Step — Open browser

Go to

```
http://localhost:8501
```

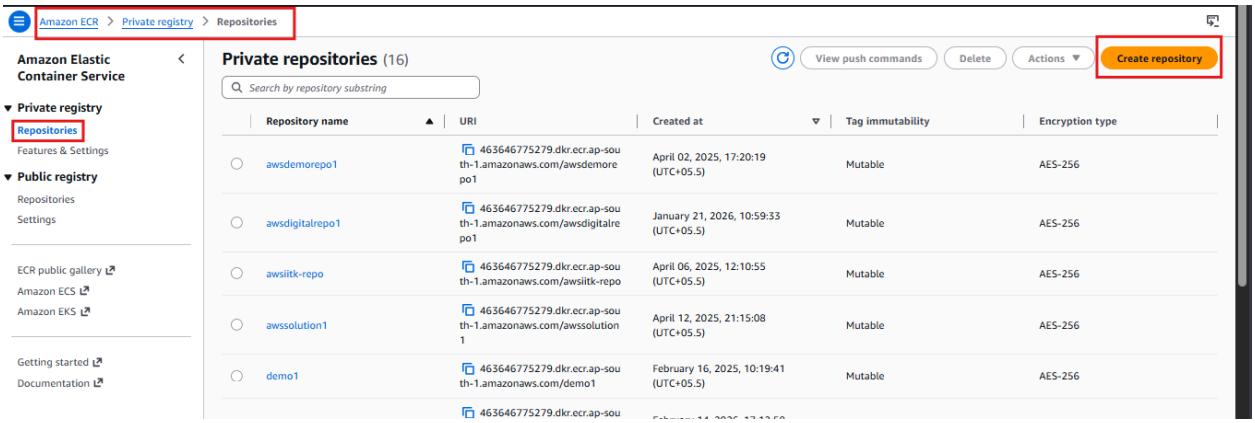
Step — Verify dashboard

You should see 3 tables with metrics

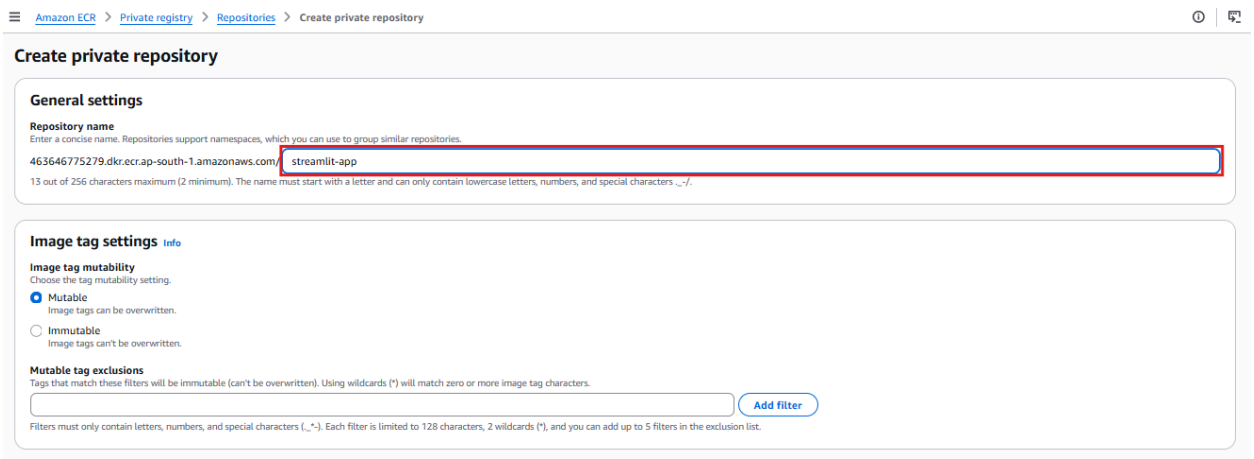
Task 9 — Push Image to ECR

Step — Create repository

ECR → Create repository → create repository



Name it : `streamlit-app` —→ Create



Step — Tag image

```
docker tag mobile-signal-app <repo-uri>
```

Step — Login

```
aws ecr get-login-password
```

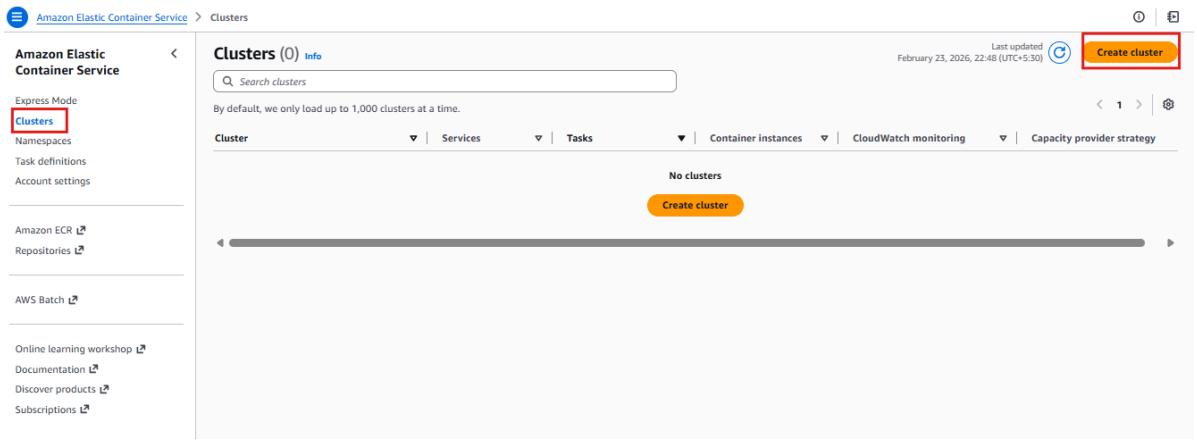
Step — Push image

```
docker push <repo-uri>
```

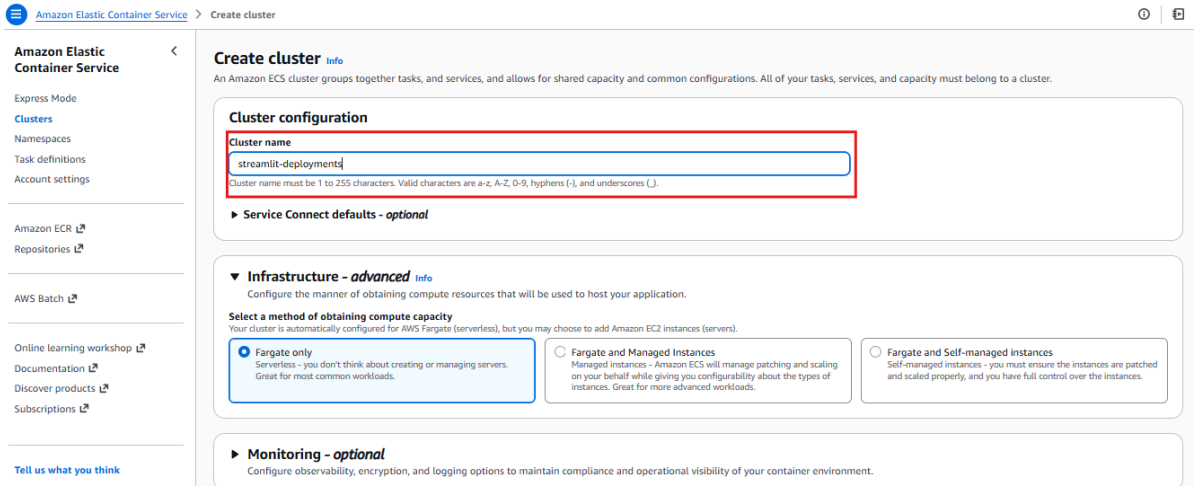
Task 10 — Deploy App to ECS

Step — Create Cluster

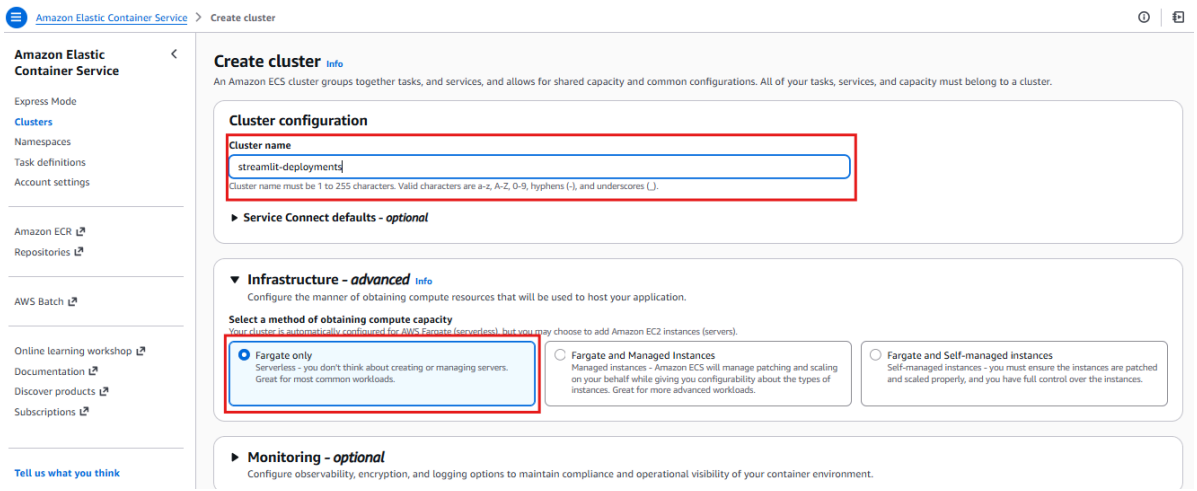
- ECS → Create cluster



- Cluster name: Streamlit-deployments

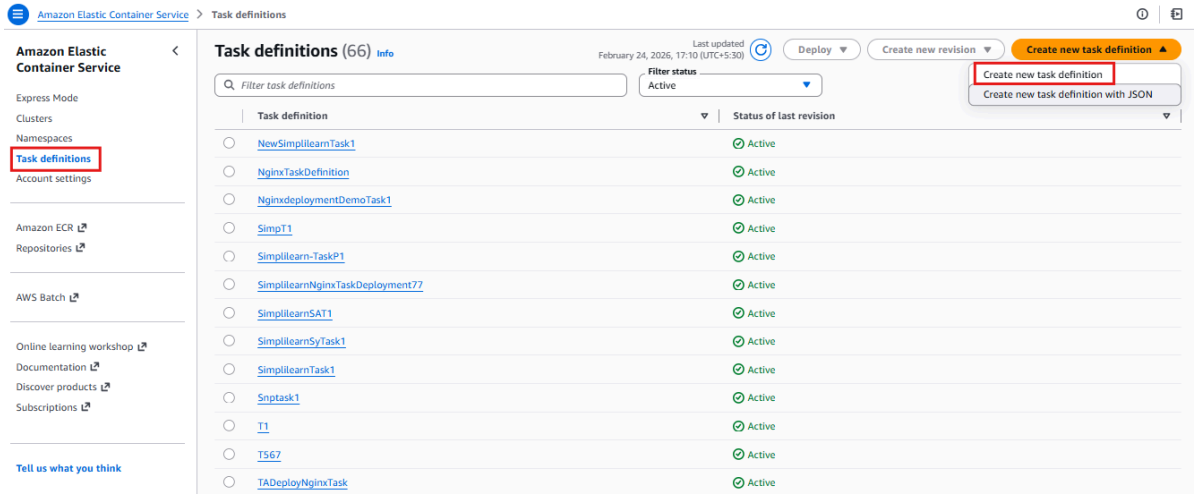


- Infrastructure —→ Fargate only —→ Click Create

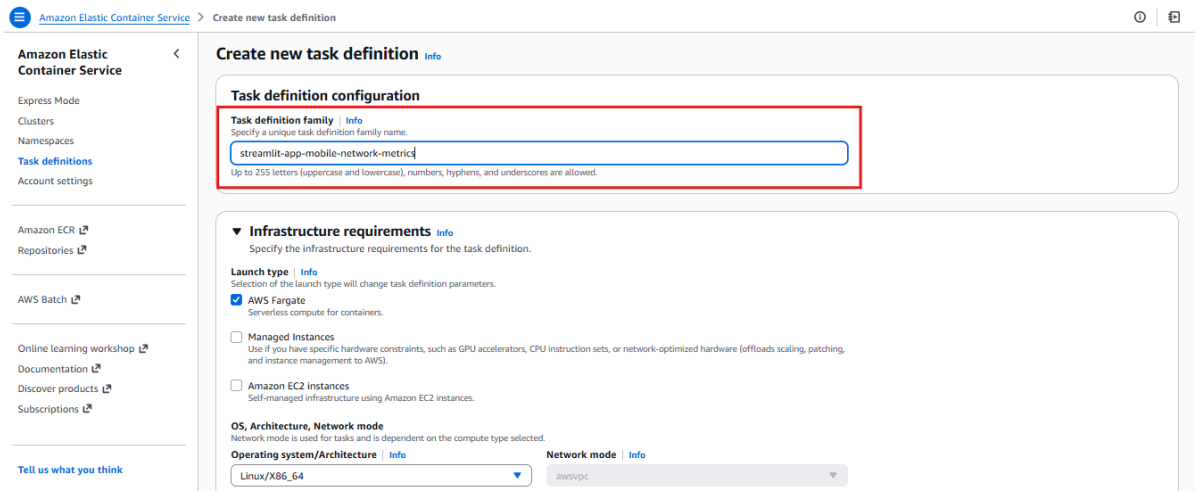


Step — Create Task Definition

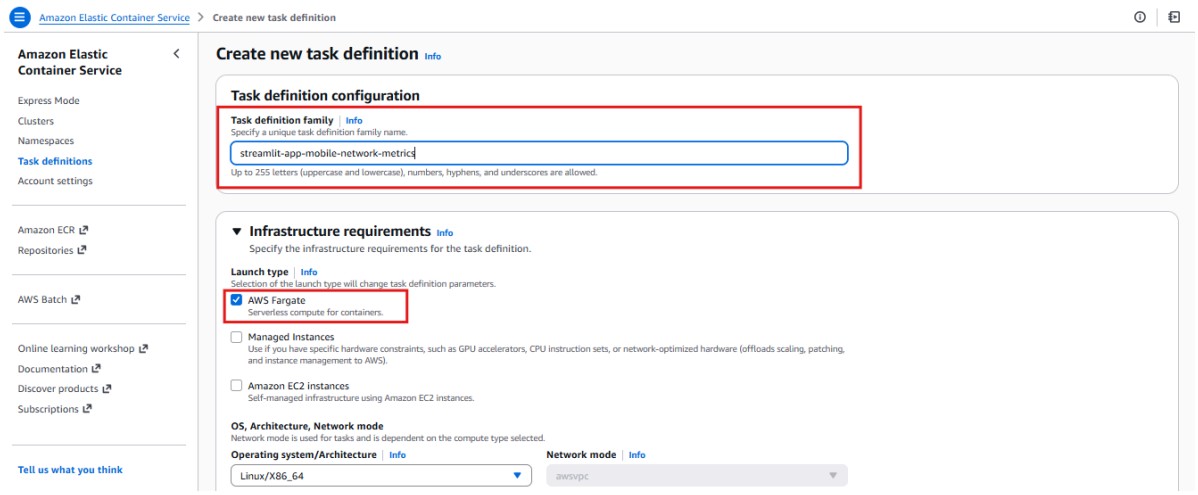
- Click on Task definition —→ Create new task definition



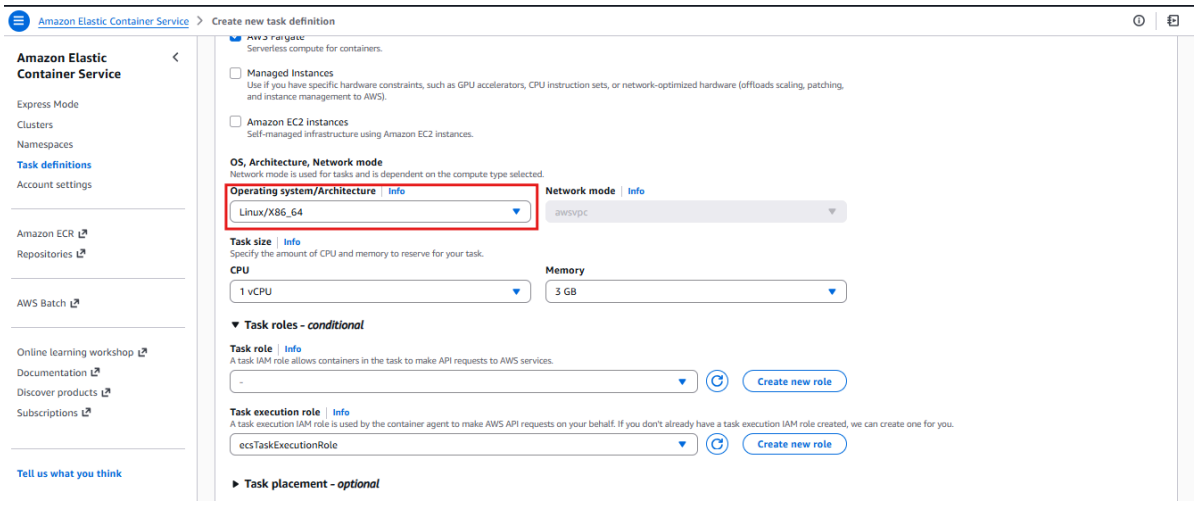
- Give it a name



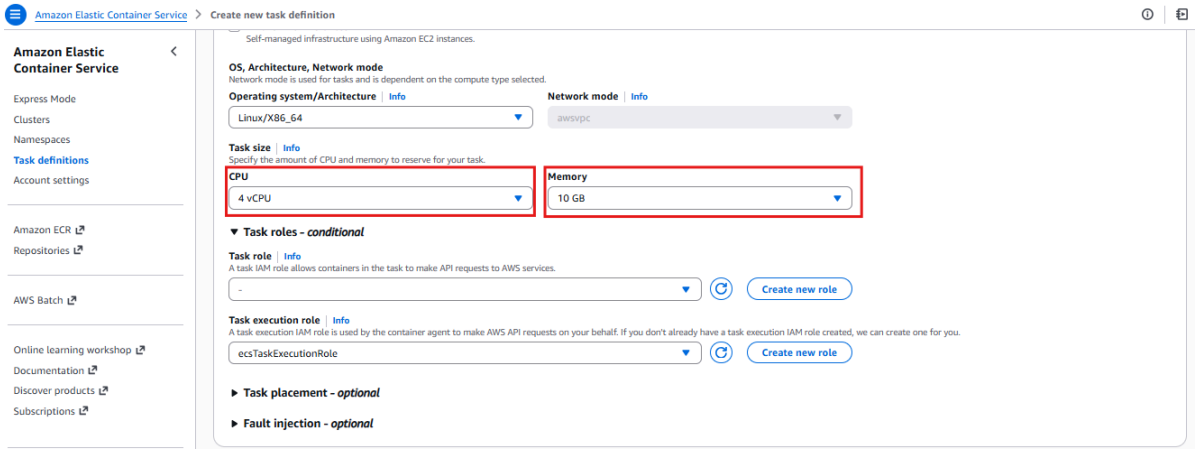
- Select Launch type → Fargate



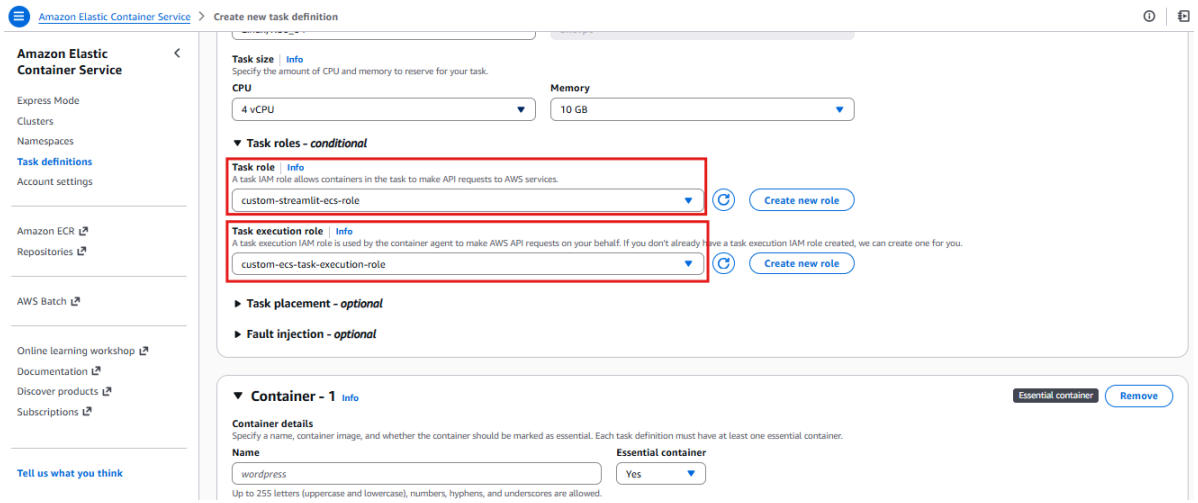
- OS → for as per your system (for win: Linux/X86_64 & for mac : Linux/ARM64)



- Select CPU → 4 vCPU ——→ Memory → 10GB

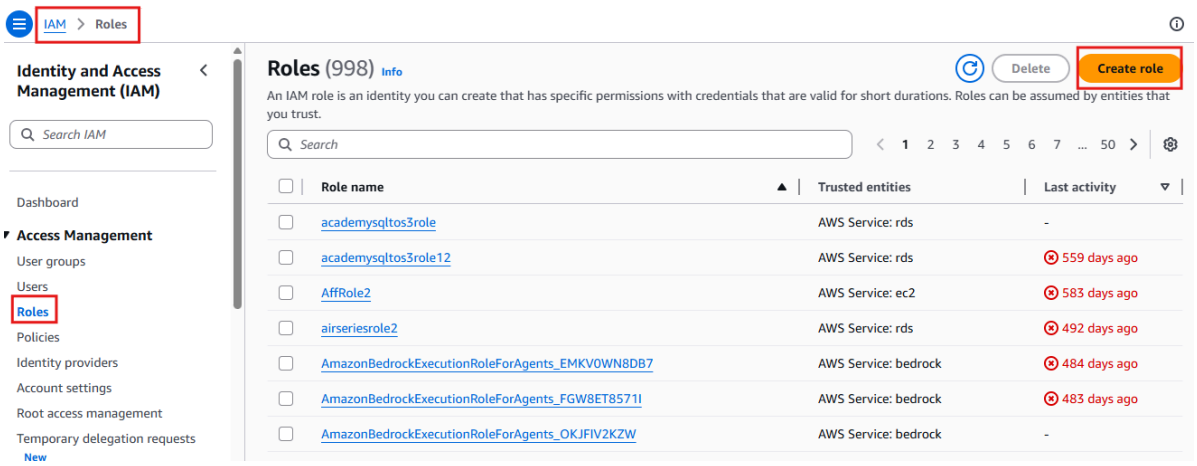


- Select IAM role —→ Go to IAM create role —→ Select the created role

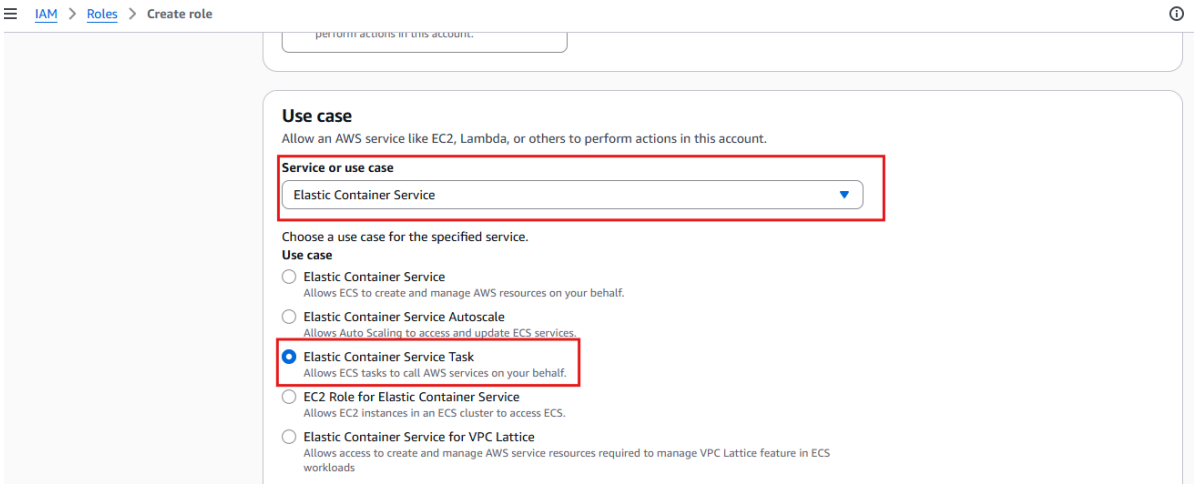


Step — Create IAM Role

Navigate to IAM —→ roles —→ create role



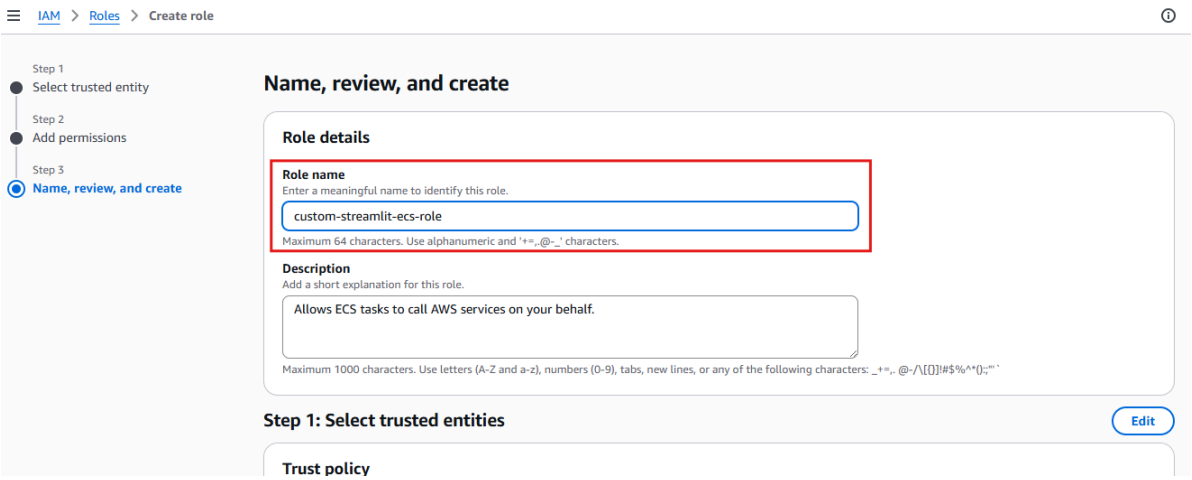
Select use case —→ Elastic Container Service —→ Elastic Container Service Task



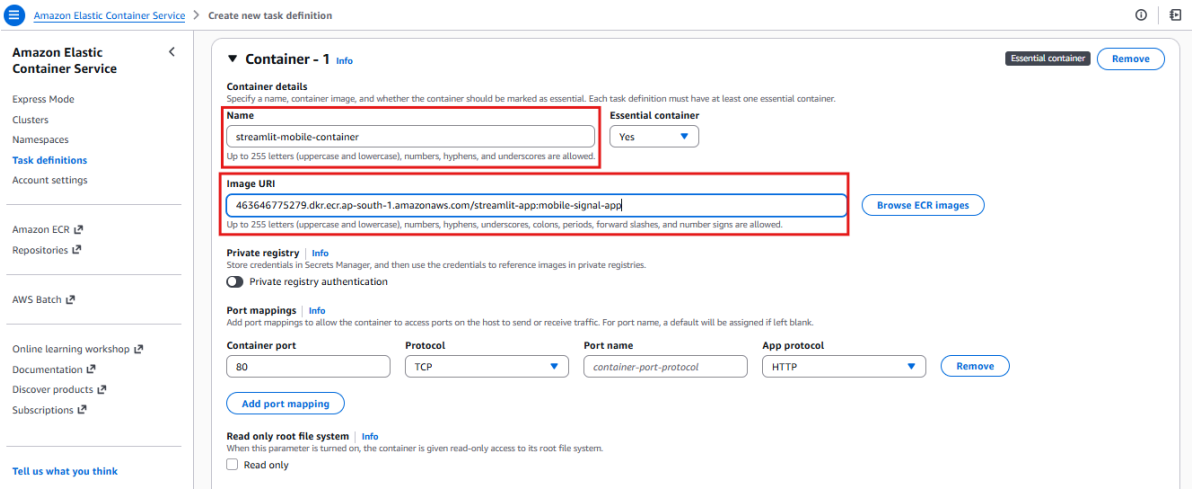
Attach policies:

- Athena Full Access
- ECS Task Execution
- S3 Full Access
- CloudWatch Logs

Name the role —→ Click creat



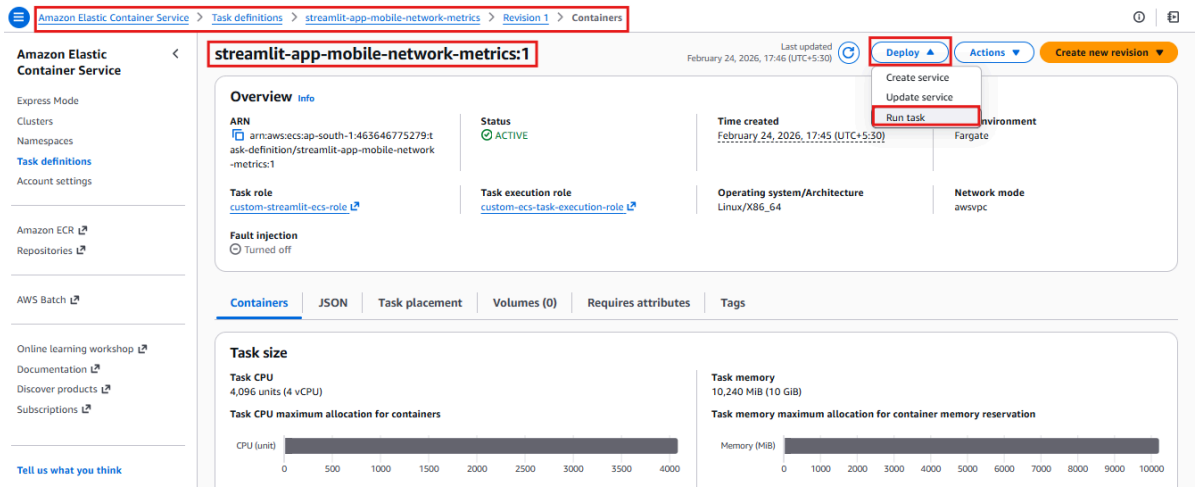
- Give container name —→ Image URI → Your ECR image URI



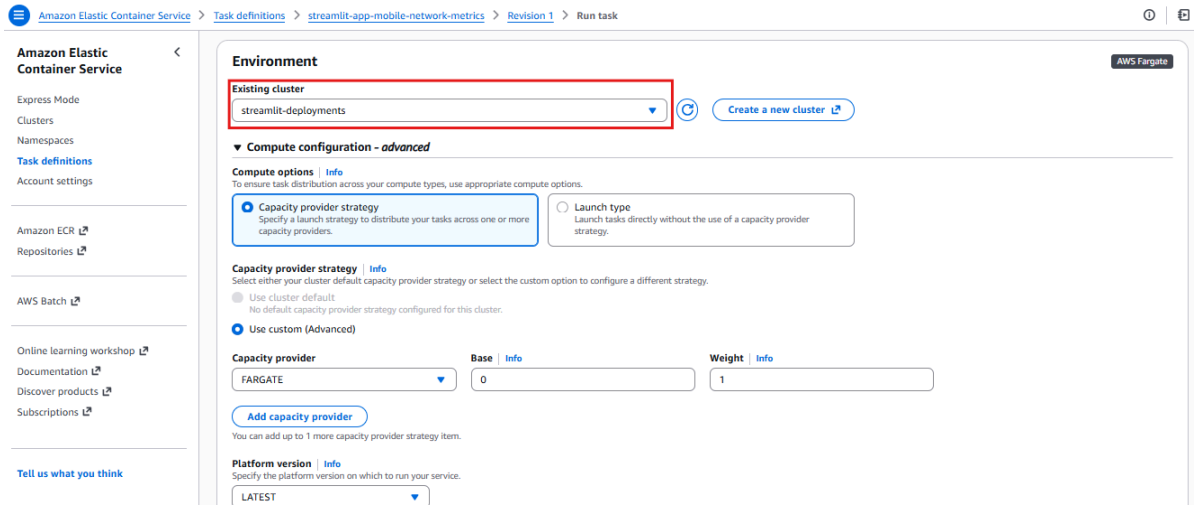
- Click on create

Step — Run Task

1. Click Deploy → Run task



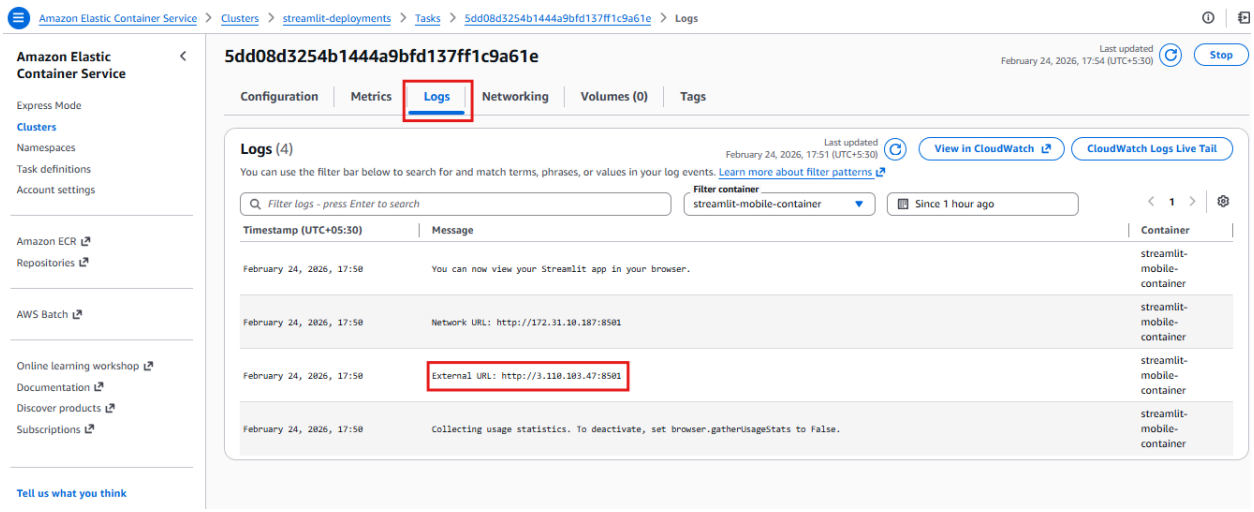
2. Select cluster



3. Leave other values as default —→ Click create

Step — Open app

Copy external URL from logs



Output

At the end of this lab you will have:

✅ Real-time streaming pipeline

✅ Spark streaming transformations running on Glue

✅ Partitioned parquet datasets in S3

✅ Glue Data Catalog tables

✅ Athena queries working

✅ Streamlit dashboard deployed on ECS

✅ Near real-time telecom metrics visualization